

Particulate Matter

Week in Review

Week ending 15 August 2026 | 9 August 2026 to 15 August 2026 | 134 records pooled from 7 daily issues

134

RECORDS POOLED
(134 UNIQUE DOIs)

45%

ON THE MEASURE-
MENT
SIDE (60 RECORDS)

10

SUBTOPICS
ACTIVE

32

POOLED EFFECT
ESTIMATES

0

NEW TRIAL
REGISTRATIONS

The week in one paragraph

The measurement side gave ground this week — 45% against 54% — and what replaced it was clinical-endpoint work in organ systems the watch had not seen: kidney, thyroid, brain white matter, Parkinson's. The instrument literature meanwhile turned from where a calibration fails to *how long* a sensor survives, which is the same question one level down.

Last week's finding was that four groups independently stopped publishing new low-cost sensor correction equations and started publishing the failure boundaries of existing ones. This week that inquiry got more specific and more uncomfortable. **Benjamin et al.** (*Sensors*) show that a low-cost CO and PM_{2.5} unit calibrated in one environmental regime does not extrapolate to another — the failure is in the environment, not the reference. **Balasubramanian et al.** fit a PurpleAir correction across the whole United States, which is the opposite move and is useful precisely as a contrast: a national model is a statement that regional structure can be absorbed, and Benjamin's result is a statement that it cannot. **Fawaz et al.** address a failure mode the correction literature does not touch at all — surface fouling of a SAW microparticle sensor in a cascade impactor, cleaned in situ by SAW-induced droplet displacement — and **Kirchhoff et al.** re-examine the working fluid in condensation particle counters. Two of those four are about the sensor surviving contact with real air, not about fitting its output.

On the health side the week's five tier-A clinical records land in five different organ systems: **Mandal** (eGFR decline, 12,271 Indian adults), **Wang et al.** (thyroid, $n = 24,333$), **Sullivan** (childhood white-matter microstructure), **Delaney** (first Parkinson's hospitalisation) and **Chow** (blood metabolomic signatures linking air pollution to lung cancer). **Not one of the five measured the exposure it analysed.** All five inherit a modelled surface — satellite, land-use regression or a spatiotemporal product — and none reports the model's error at the concentrations where the effect is claimed. **Across 134 records and 34 sensing papers, the count of records that validated an instrument against a health endpoint is again zero,** for the third consecutive weekly rollup.

Subtopic trend and inflection

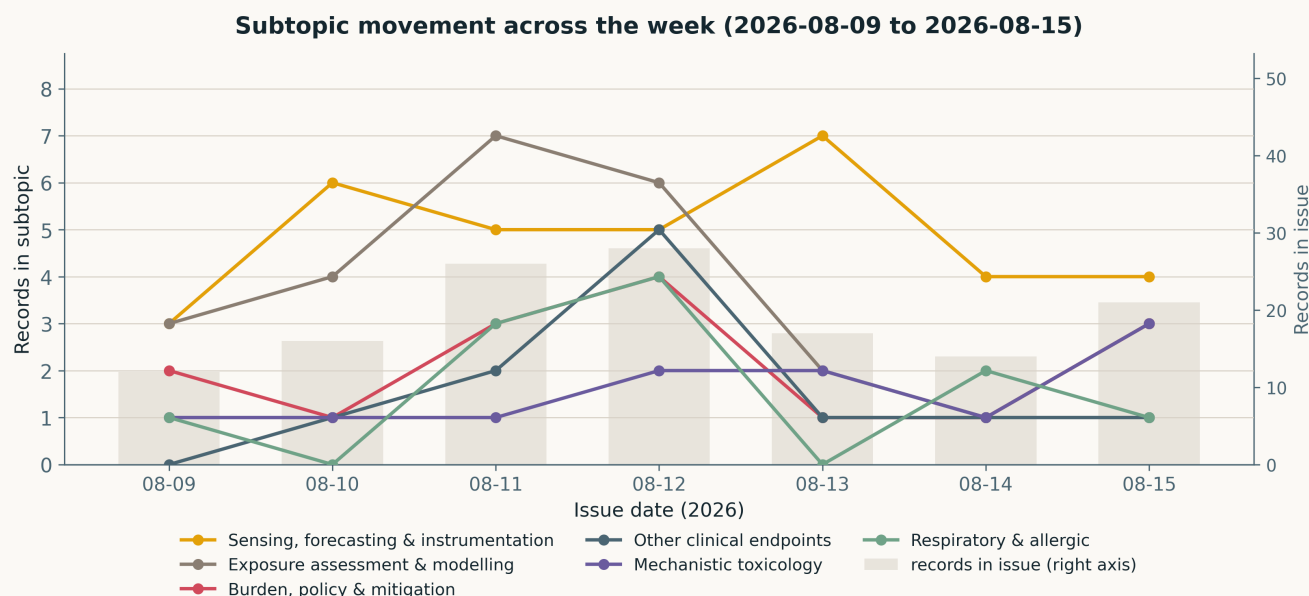


Fig. Per-issue subtopic counts across the week from `state/metrics.csv`. **Read the denominators first:** daily totals were 12, 16, 26, 28, 17, 14 and 21 — a factor of 2.3 between the largest and smallest day, and a much flatter week than 2–8 August, whose range was 10 to 31. The 11–12 August peak (26 and 28) is an Elsevier *Atmospheric Environment* deposit batch, visible in the register as nine `atmosenv.2026.1222xx/1223xx` DOIs arriving across two days, most without abstracts. **That is a publisher's release schedule, not the field.** The 9 August floor of 12 is a Sunday. Once those two artefacts are removed the week has no inflection at all, and the honest summary is that subtopic composition was stable and the total declined slightly.

Fading, and the one number worth carrying forward. Sensing dropped 47 → 34 and exposure modelling 32 → 26; the measurement-side share fell from 54% to 45%. Both of last week's peaks were partly this pipeline's own doing — the Crossref-by-ISSN leg was widened on 2 August and immediately surfaced ten calibration papers — so the fall is at least partly a return to baseline rather than a decline in the field. **Do not read —13 as a signal.**

Emerging. *Clinical endpoints outside the cardiorespiratory core.* Other clinical endpoints rose 6 → 11 and Neuro / mental health 2 → 5, and the content is genuinely new for this corpus: renal function (Mandal; Tan et al. on eGFR with nutritional and activity indices), thyroid (Wang et al., $n = 24,333$; a nodular-thyroid record), haematological malignancy and CHIP, and Alzheimer's neuropathology. **This is the first week in which the watch's clinical tail was longer than its cardiorespiratory trunk,** and it is the shift most likely to persist, because it tracks the arrival of large linked-registry cohorts rather than a publication batch.

Flat but load-bearing. Mechanistic toxicology held at exactly 11 for the second week, occupational and indoor at 9, cardiovascular at 7. Three clusters unchanged to the record across a fourteen-day window is worth stating: this corpus has a stable substrate and a volatile surface, and almost all week-to-week movement is in sensing, exposure modelling and burden — the three clusters most sensitive to which harvesting leg happened to fire.

Study architecture

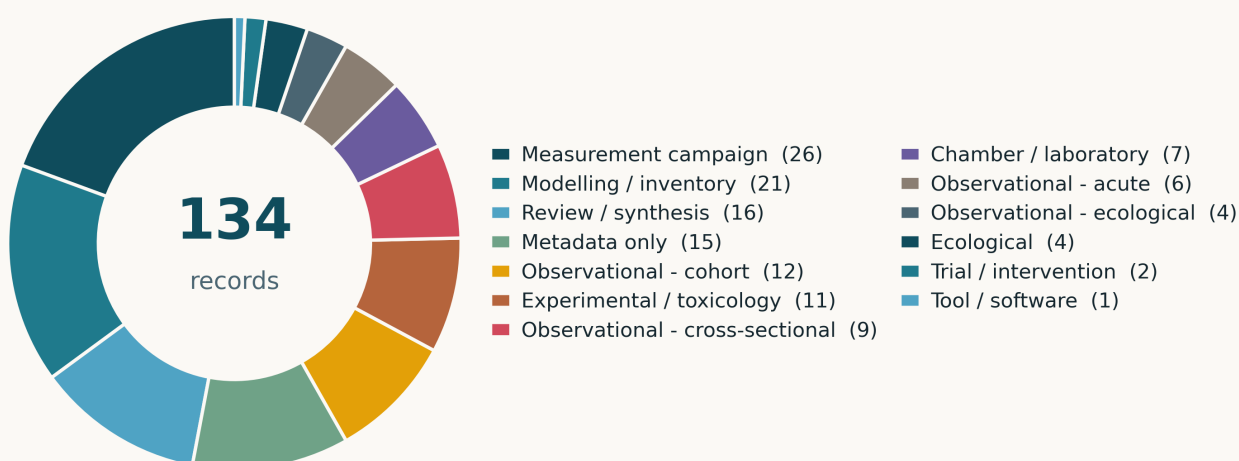


Fig. Pooled study architecture over 134 records. Measurement campaigns (26) and modelling/inventory (21) lead, as in every week of this watch. **Two slices should be read as defects rather than designs.** *Metadata only* is 15 records — 11% of the week — surfaced with no abstract in Crossref, Europe PMC or Semantic Scholar and therefore classified but unread; nine of the fifteen are the *Atmospheric Environment* batch above. *Review / synthesis* is 16 and is a publication-cycle artefact, not new evidence. **The Trial / intervention slice is 2 records in 134** — a protocol and a feasibility pilot, neither reporting an outcome. **For the third week running, not one completed randomised intervention on a PM exposure entered the corpus,** while 21 such trials sit in the registry watch below.

Pooled analytics

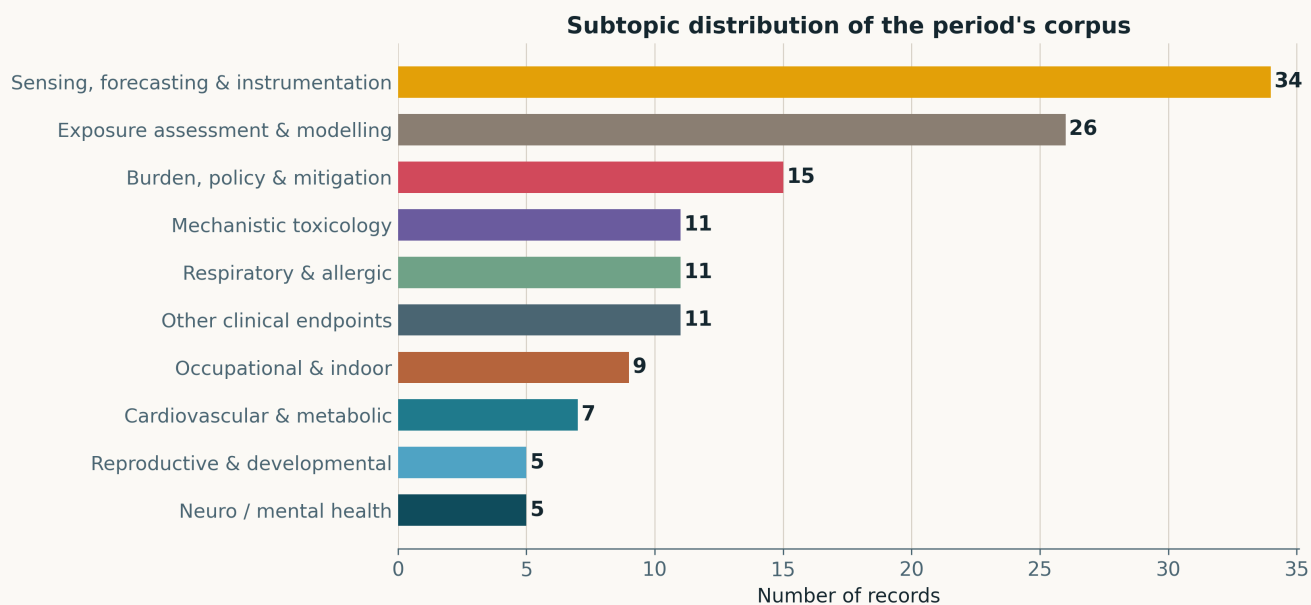


Fig. Pooled subtopic distribution, 9 August 2026 to 15 August 2026. **Sensing (34) and Exposure assessment & modelling (26) account for 60 of 134 records, 45% of the week**, down nine points on the preceding week. All ten canonical clusters were active, which is a property of a seven-day window rather than of any day — *Neuro / mental health* was empty on five of the seven issues and still finished at 5. The clinical clusters now form a genuine block rather than a tail: *Other clinical endpoints* (11), *Respiratory & allergic* (11), *Occupational & indoor* (9), *Cardiovascular & metabolic* (7), *Reproductive & developmental* (5) and *Neuro* (5) total 48 records against 60 on the measurement side.

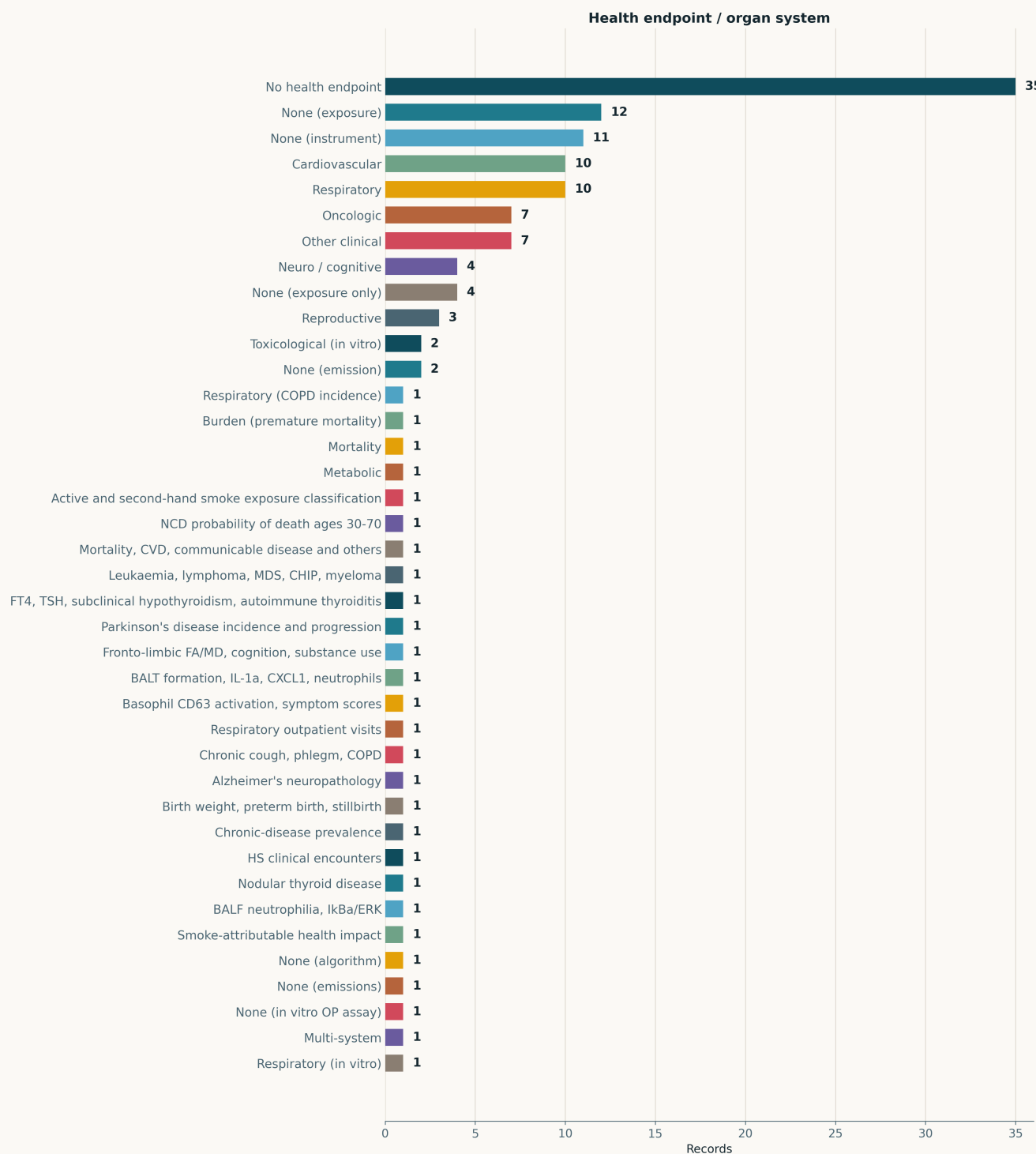


Fig. Pooled health endpoint by organ system. **The chart is misleading as drawn and the reason is a label defect, not a finding.** “No health endpoint” shows 35, but seven further free-text labels — None (exposure) 12, None (instrument) 11, None (exposure only) 4, None (emission) 2, and one each of None (emissions), None (algorithm) and None (in vitro OP assay) — mean the same thing and are not in ENDPOINT_CANON. **The true no-endpoint count is 67 of 134, or 50%,** which is the direct consequence of the 45% measurement weighting plus the 15 unread metadata-only records. Logged for repair in `plots.py`. Of the records that do carry a human endpoint, respiratory (10) and cardiovascular (10) lead, with oncologic at 7 — the highest oncology count this watch has recorded. **Every bar below 3 records is one or two studies; do not read the tail.**

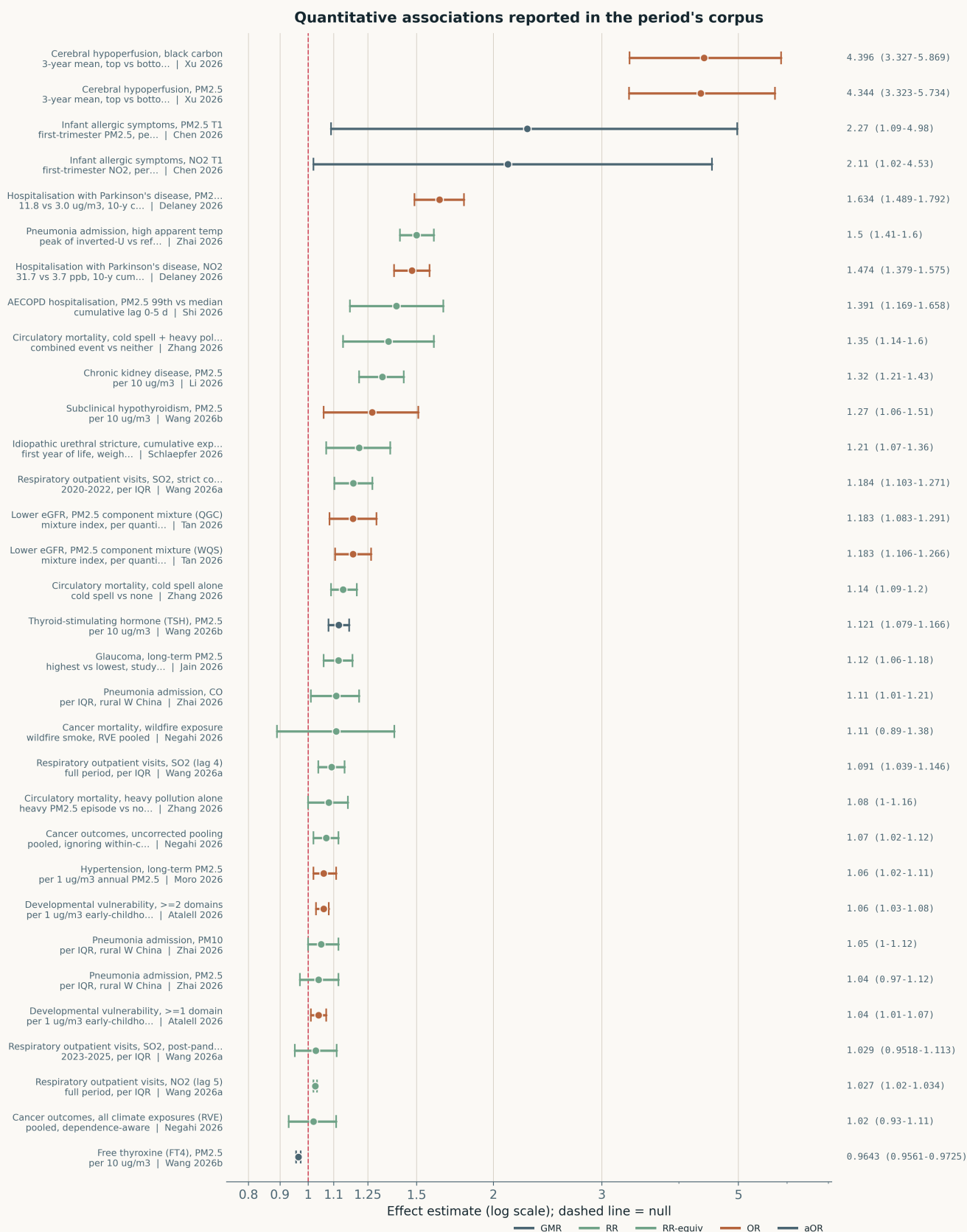


Fig. Pooled effect estimates for the week — a caterpillar plot, not a meta-analysis. No summary diamond is drawn and none should be. Pooling is not defensible and no I^2 is reported, for the same four independent reasons as last week. (i) *Exposure contrasts differ*: per $1 \mu\text{g}/\text{m}^3$ (Moro, Atalell), per IQR, per $10 \mu\text{g}/\text{m}^3$, 99th-percentile-versus-median at cumulative lag (Shi), and whole-event contrasts (Zhang's cold-spell/pollution scenarios). (ii) *Estimands differ*: 14 RR, 10 OR, 4 RR-equivalent, 2 GMR and 2 adjusted OR share an axis but not an interpretation. (iii) *Records are not independent* — Zhang contributes three nested contrasts from one meta-analysis, Atalell two nested domain thresholds from one cohort, Negahi three estimates two of which are the same data under different dependence assumptions. (iv) *Not all exposures are PM*. The most informative single fact on this plot is the width distribution, not the location: the tightest intervals belong to whole-of-population registry cohorts with per-unit contrasts near the null, and the widest to small cohorts with large contrasts. That is a statement about study size, and reading the ordering as an effect-size ranking would be an error.

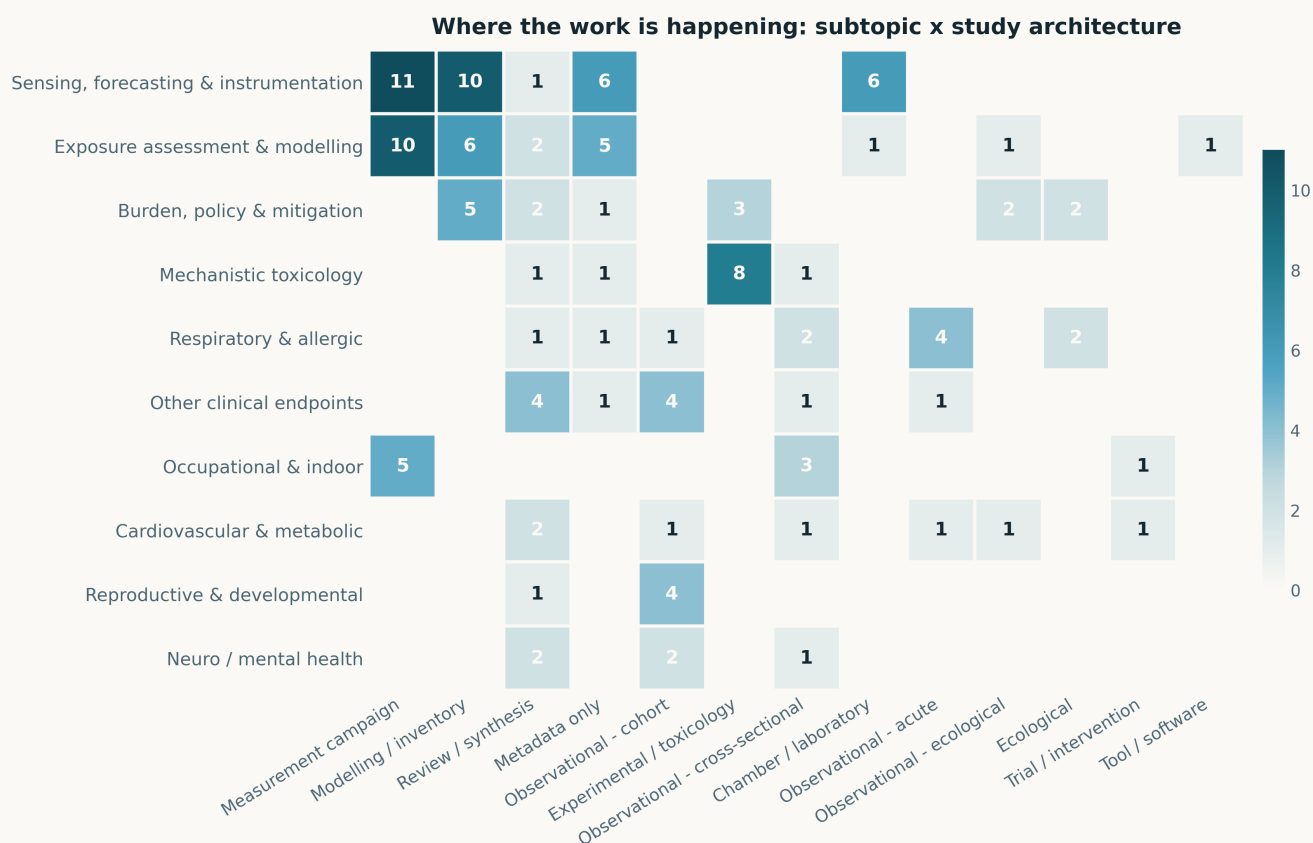


Fig. Pooled subtopic $\times$ study-architecture cross-tabulation over the full week. 134 records occupy 47 of the 130 available cells. The cells that stay empty are the point. At $n = 134$ the *Sensing* and *Exposure assessment* rows still contain no *Trial / intervention* cell and no cell coupling an instrument to a health outcome — the third consecutive week this has held. *Mechanistic toxicology* remains almost entirely experimental with no bridge to human biomarker work, unchanged since 27 July. Three weeks of accumulation and 575 records since 26 July, and the coupling cell has never been occupied: at this point it is a property of how the field is organised, not of sampling.

Groups, venues and most-cited work

Recurring author and group clusters. The naive first-author count is **misleading and must not be used**: the top entries are Liu (6), Zhang (6), Wang (4), Chen (4), Li (3) and Wu (3), which are among the most common surnames in the world and almost certainly denote distinct groups — inspection of affiliations finds no two of the six Zhang records sharing an institution. **One genuine repeat survives**: Cho appears twice, on the Norwood community-science mobile campaign (15 August) and earlier in the week, and those are the same programme. **No institution contributed more than two records**, and there is no evidence of a single programmatic output stream inflating apparent consistency. **Deduplication was clean this week: 134 records over 134 unique DOIs**, against 146 over 145 last week — the cross-issue duplicate that shipped on 2 and 3 August has no counterpart in this period.

A label defect repaired by this rollout. Six records carried on 11 August were stored with a journal name in the author field (Atmos Environ 2026a-d, Aerosol Air Qual Res 2026, Atmos Pollut Res 2026) because the Elsevier and Springer deposits carried no abstract and the daily run did not resolve authorship. A Crossref re-query during this rollout resolved all six — Zhou, Habib, Hao, Li, Hsieh and Yavuz — and the corpus store has been corrected. **The 11 August PDF as shipped still shows the journal names in its register**; the store, this rollout and every future rollout show the authors. Recorded rather than silently fixed.

Venue concentration. *Atmospheric Environment* is the largest single contributor at **13 records**, and **that is one publisher deposit batch across 11–12 August, not a research surge** — nine of the thirteen arrived on two consecutive days without abstracts. After it: *Environmental Monitoring and Assessment* 7, *Environmental Pollution* 6, then *Journal of Hazardous Materials*, *Atmospheric Chemistry and Physics*, *Environmental Science & Technology*, *Journal of Thoracic Disease*, *Atmospheric Measurement Techniques*, *Sensors* and *Research Square* at 4 each. **The free-text venue defect recorded last week persists**: Research Square (preprint) (4) and Res Sq (preprint) (3) are the same venue under two spellings, so preprints are actually the second-largest source of the week at 7 records.

Most-cited work of the period. Not reportable, and ranking it would be noise. Every record entered a PubMed, Europe PMC or Crossref index within the last seven days; citation counts are zero or near-zero across all 134 and carry no signal about importance. A defensible most-cited list requires a lag of at least 18 months and will be reported in the yearly issue, not weekly.

Trial registry watch

From `state/trials.json`. Daily issues accumulate; the rollout is where they surface. **Zero registrations entered state during 9–15 August** — ClinicalTrials.gov was checked on each of the seven daily runs and returned no PM or air-pollution registration created or updated inside any daily window. The standing list of **21** is unchanged since 8 August. Note the hard limitation recorded previously: the connector exposes *no* last-update-date filter, so these are relevance sweeps rather than true window queries, and `first_seen` means “first seen by this pipeline”.

A zero week is itself informative, with a caveat. Seven consecutive days with no new PM-exposure registration is consistent with the structural picture this watch has built: the interventional literature in this field is small, almost entirely filtration-based, and adds a few registrations a month rather than a few a week. But the absence of an update-date filter means a *silently updated* existing registration would not surface, so read this as “no new trial found”, not “no registry activity”.

Results now overdue — the list has grown. **NCT05994937** (NYU Langone, portable air cleaner, glycaemic and cardiometabolic biomarkers, $n = 150$) reached primary completion **2026-03-16** with no results posted — now **five months overdue**, and note that its protocol was published this week as a daily record (Wittkopp et al., *Am Heart J*, 14 August) while the results remain unposted, which is a reporting-order worth flagging. **NCT06534385** (Alaska Native Tribal Health Consortium, $n = 15$, 2026-04-01), **NCT07047430** (U. Washington, airport-adjacent asthma, $n = 60$, 2026-06-01) and **NCT07111208** (MARKOPOLO, noise and ultrafine particles, $n = 180$, 2026-07) are all past primary completion with nothing posted. **NCT07479420** (cluster-randomised HEPA versus ionisation/plasma air cleaners, $n = 180$) is marked COMPLETED and is the single registration in state most likely to produce a directly usable exposure-reduction estimate; it should be watched weekly.

Publication-risk item. **NCT06197477** (Hasselt, school air purifiers, $n = 900$ — the largest trial in state) remains UNKNOWN past a December 2025 primary completion, with four cognitive co-primary endpoints and no stated multiplicity control. Eight months of unknown status on the largest registered trial in the field is the single biggest outstanding gap in the interventional evidence base.

Adequately powered and still reading out. **NCT06376994** (JHSPH, air cleaner versus sham in COPD, $n = 770$, 2029-08-31) and **NCT05874479** (NYU Langone, blood pressure in NYC public housing, $n = 440$, 2027-09) remain the only two registrations that can plausibly detect a modest effect on a surrogate endpoint. Neither reads out inside two years.

Open gaps

- ▶ **Measurement–epidemiology coupling: zero for a third week.** Across 134 records and 34 sensing papers, **not one validated an instrument against a health endpoint.** The nearest approach this week is Makacha et al. (14 August), which ran minute-resolved *personal* PM_{2.5} monitoring on 343 women against facility reference monitors and found ambient–personal concordance of **0.01–0.56 with slopes below 1 in every setting** — an exposure-misclassification measurement with no health outcome attached. Put it beside the five tier-A clinical records above, none of which measured its own exposure, and the shape of the gap is exact.
- ▶ **Every large clinical result this week rests on an unvalidated exposure surface.** Mandal (1 km spatiotemporal model, Delhi annual means 118–130 µg/m³), Moro (satellite model over rural Ghana), Atalell (SA2-level daily PM_{2.5} at a 1 µg/m³ contrast), Sullivan and Delaney (modelled residential surfaces). **None reports out-of-sample model error at the exposure levels where the effect is claimed.** This is not a criticism of any one paper — it is the field convention — but at per-unit odds ratios of 1.04–1.06 the convention is now the binding constraint on inference.
- ▶ **Ultrafine and speciated metrics: worse than last week.** Of 134 records, **exactly one** classifies as ultrafine/nanoscale and four as size-resolved or number distribution, against 48 PM_{2.5}-only and 16 joint PM_{2.5}/PM₁₀. Composition and speciation reached 8, and optical properties 7 — both up, and both driven by the *Atmospheric Environment* deposit batch rather than by sustained interest. The deposition physics that determines dose remains the most under-served measurement axis in this corpus.
- ▶ **Geographic coverage against attributable burden: unchanged and still inverted.** China 31, North America 20, Europe 19 against **South Asia 6 and sub-Saharan Africa 5.** South Asia and sub-Saharan Africa together carry a far larger share of global PM-attributable DALYs than China, North America and Europe combined, and contributed 11 of 134 records. Two of those 11 — Mandal and Moro — are this week's strongest clinical records, which makes the scarcity more striking, not less.
- ▶ **Unread records are 11% of the corpus.** Fifteen records carry no abstract from any checked source and were classified but not evaluated; a Crossref re-query during this rollup resolved **0 of 15** abstracts, though it did resolve six authorships. The nine *Atmospheric Environment* records among them are exactly the instrumentation and source-apportionment work this watch most wants to read. **Absence of criticism in the register does not mean a record is sound.**
- ▶ **Negative and null results: present, which is a good sign.** Moro reports a clean null for metabolic syndrome and all its non-hypertension components alongside a positive hypertension estimate; Hung reports no significant mediation through any measured inflammatory cytokine, against the default assumption in that literature; Atalell reports null prenatal and cumulative windows alongside a positive early-childhood window. A corpus with no nulls is a reporting-bias signal. This one has them and they are load-bearing.

Where this is heading — judgement, clearly marked as such

The instrumentation literature is descending the reliability stack, and that is the most useful thing it has done in three weeks. Two weeks ago the question was whether a correction fits. Last week it was where a correction stops working. This week the questions were sensor surface fouling (Fawaz), working-fluid choice in a CPC (Kirchhoff), environmental extrapolation limits (Benjamin) and deployment-scale correction structure (Balasubramanian). Those are maintenance, drift and physical-integrity questions — the things that actually determine whether a five-year network deployment yields usable data, and the things a held-out R^2 on a two-week co-location cannot address. Confidence: moderate. Four papers over seven days is a thin base, and the same caution applies as last week — some of what looks like a turn is this pipeline's harvesting legs firing differently.

What would change the picture. The same single paper as last week: a calibrated low-cost network reporting an exposure–response against a clinical endpoint. Nothing in the 21 registered trials will supply it. The likelier route is now visible in this week's corpus — a large linked-registry cohort (the kind producing the thyroid, renal and Parkinson's results) retrofitting a dense sensor layer onto outcome ascertainment it already has. That requires no new trial, only a cohort willing to instrument its participants' addresses.

What a distributed low-cost PM sensing programme should do differently. Three things this week's corpus argues for, and they are cheaper than the alternative. (1) Budget for *maintenance* as a measured quantity, not an operational afterthought: Fawaz's in-situ cleaning and Benjamin's extrapolation limits both say node performance degrades in ways a deployment-time calibration does not capture, so schedule re-co-location and report performance as a function of time since it. (2) Report exposure-surface error at the concentrations of interest, because every large clinical result this week would have been strengthened by one number nobody supplied. (3) Site the next network where the burden is, not where the reference monitors are — South Asia and sub-Saharan Africa produced this week's two best clinical results on the worst exposure data in the corpus, and that is the highest-leverage measurement gap available. **The week was quieter than the last two, and more clinically diverse than any week so far.**

Paper-level digest — the week by subtopic

Every record of the period, grouped by subtopic. **Two tiers, and the difference matters.** Entries carrying a full critical summary are lifted *verbatim* from the archived daily issue that first published them — they were read, and the stated weaknesses are assessed weaknesses. The remainder appear as precis or metadata-only entries in the lighter style: those records were surfaced and classified but **not read**, either because no abstract was available in Crossref, Europe PMC or Semantic Scholar (15 records this week, 11% of the corpus) or because they entered on metadata alone. **Absence of criticism in the lighter tier means the record is unassessed, not that it is sound.** Do not cite a precis entry as though it had been evaluated.

Sensing, forecasting & instrumentation

34 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (34 OF 34 RECORDS IN THIS SUBTOPIC)

Acton-Bond et al. 2026 Atmos Environ

Rolling positive matrix factorisation for aerosol optical apportionment

METADATA ONLY — no abstract in Crossref, Europe PMC or Scholar Gateway. Listed, not summarised. Carried because *rolling* PMF applied to optical rather than chemical data is a specific and consequential claim: conventional PMF source apportionment requires filter sampling and offline speciation, whereas an optical implementation with a rolling window could in principle run on an aethalometer or a multi-wavelength absorption photometer at deployment cadence. That would move source attribution from a laboratory capability to a network capability, which is the distinction that decides whether a distributed sensing programme can report anything beyond mass. Whether the paper supports that reading is unknown until an abstract appears.

doi: 10.1016/j.atmosenv.2026.122295

Adams et al. 2026 Environ Sci Atmos

Sampling sub-pixel scale variability in total column NO₂ within a coastal urban environment

Long-term Pandora spectrometer records of column-integrated NO₂ at Boston University and Harvard University, used to characterise variability within a single satellite pixel. Directly relevant to any satellite–surface fusion product: sub-pixel heterogeneity is an irreducible error term in the validation of column retrievals, and two sites a few kilometres apart in the same coastal urban area are the cleanest way to bound it. Two instruments in one metropolitan area is a small basis for a general claim about pixel-scale variability, and NO₂ column heterogeneity is not a direct proxy for PM heterogeneity.

doi: 10.1039/d6ea00033a

Akazezuwa et al. 2026 Environ Monit Assess

SFFO-RL intelligent environmental monitoring: smart sensing networks with artificial intelligence

Proposes an “Intelligent Environmental Monitoring Framework” combining adaptive filtering, Isolation Forest outlier removal, min–max normalisation, multi-order neighbour feature fusion and a reinforcement-learning optimiser, reporting **97.1 % accuracy, 97.8 % F1, 97.6 % recall and 98.21 % precision**. Those figures cannot be interpreted. The abstract names no pollutant, no dataset, no sensor, no units, no sample size and no held-out split; “accuracy” is quoted for what is elsewhere described as predicting continuous pollutant concentrations, which is a category error unless the task was silently discretised. A four-metric cluster in the 97–98 % band on environmental time series with autocorrelation is the signature of evaluating on temporally adjacent data. Carried as tier C because the framework class (edge-side denoising plus learned correction) is squarely on this watch’s axis and the record should be visible in the register; the reported numbers should not be cited.

doi: 10.1007/s10661-026-15724-0

Albashir 2026 Res Sq (preprint)

A physics-informed digital twin framework for health vigilance and indoor air quality decision support

Preprint proposing a physics-informed digital twin for indoor air quality in healthcare-oriented environments, arguing that monitoring-only systems do not support proactive decisions. **Not peer reviewed, single author, no deployment, no validation against measured concentrations reported in the abstract.** Carried only because the architectural claim — coupling a physical model of a room to a sensor stream so the system can extrapolate beyond sensor locations — is the same idea as physics-informed calibration on the ambient side, and the indoor literature is where it is being tried first. Treat as a pointer to a design pattern, not as evidence. Relevance to distributed sensing: physics-informed correction is the leading candidate for generalising calibration away from co-location sites, and this is a low-cost read on how that framing is developing indoors.

doi: 10.21203/rs.3.rs-10034990/v1

Alexa et al. 2026 Talanta

Simultaneous continuous determination of gaseous HNO₃ and particulate nitrate by chemiluminescence

Gaseous HNO₃ is sampled by wet diffusion denuder and particulate NO₃[−] in parallel by continuous aerosol sampler, both analysed online by continuous flow analysers with chemiluminescence detection — no preconcentration step, signal recorded at 1 s intervals. Detection limits (3S/N) are **5.1 nM and 12.8 nM in solution, corresponding to 0.13 μg/m³ (49.7 ppt) for HNO₃ and 0.10 μg/m³ for NO₃[−]**. Applied to urban air in Brno across winter and summer campaigns 2022–2023 and compared against filter sampling with NaF-coated dry denuders. The gas–particle partitioning of nitrate is temperature- and humidity-driven on timescales of minutes, which filter-based methods average away entirely and optical low-cost sensors cannot resolve at all; a method at this time resolution is the reference any nitrate-aware correction scheme would need. The abstract reports no field intercomparison statistics against the reference method, only that the comparison was made.

doi: 10.1016/j.talanta.2026.130457

Aparicio et al. 2026 Sensors (Basel)

Sensor system design and deployment for environmental and visitor monitoring in cultural heritage sites

Design, build and field validation of low-power low-cost nodes across five pilot sites in four European countries under the EU ARGUS project, in two configurations: static nodes at fixed positions and dynamic nodes carried by a quadruped robot or drone. Payload spans thermal–humidity probes, volumetric water content, wind, pollution monitors and two generations of visitor-counting devices, with Baltanás used as the iterative prototyping site before wider deployment. Carried for the mobile-platform architecture — a sensor-on-robot deployment pattern that is directly transferable to indoor and semi-enclosed PM mapping — rather than for any air-quality result. **The abstract reports no PM performance figures, no reference co-location and no calibration procedure**, which is the whole question for a low-cost node, so nothing here constrains data quality. Tier C on that basis.

doi: 10.3390/s26154941

Babu et al. 2026 Environ Monit Assess*Capturing the invisible: hyperlocal variability of size-resolved particulate matter in a smart city*

Mobile monitoring campaign in Madurai, Tamil Nadu, during the northeast-monsoon retreat: a vehicle-mounted size-resolved optical instrument sampling every 6 s across $\sim 2.4 \text{ km}^2$ aggregated to a $50 \text{ m} \times 50 \text{ m}$ grid. Local enhancement above route background was largest for PM_{10} at $188.91 \pm 149.07 \mu\text{g}/\text{m}^3$, attributed to road dust and construction; evening PM_{10} ($42.10 \mu\text{g}/\text{m}^3$) exceeded morning levels; the size distribution was bimodal, with coarse ($16.77\text{--}32.37 \mu\text{m}$) and submicron ($0.275\text{--}0.53 \mu\text{m}$) peaks. Getis-Ord G_i^* identified persistent hotspots on traffic corridors and cold spots in restricted-access zones; SEM-EDX on 22 elements linked size to composition. The limitation that threatens the inference is that the reported dispersion (SD 79 % of the mean) is not decomposed: on a moving platform, true spatial variance, uncalibrated optical response, inlet losses and 6 s instrument response time are confounded, and no co-location against a reference monitor is described. Directly relevant to distributed sensing: this is the spatial resolution a mobile network buys, reported without the calibration chain that would make it transferable.

doi: 10.1007/s10661-026-15746-8

Bajaj & Settle 2026 Environ Monit Assess*Rapid assessment of air pollution data quality using a composite index*

A Data Quality Index aggregating five 0–100 dimensions — accuracy (30%), calibration (20%), completeness (20%), representational consistency (15%), timeliness (15%) — with weights the authors state should vary by stakeholder. Demonstrated on hourly OpenAQ PM records for Los Angeles, 1–5 July 2025: a reference-grade monitor scored ≈ 99 and a low-cost sensor ≈ 58 , the latter penalised for calibration bias, low completeness, extreme outliers during the 4 July episode, and zero timeliness once reporting ceased on 5 July. The value is in keeping the components legible rather than collapsing to a single opaque score. The demonstration is $n = 2$ over five days and includes a fireworks episode — an outlier regime chosen, apparently, for illustration — so the numbers are a worked example, not a benchmark.

doi: 10.1007/s10661-026-15780-6

Balasubramanian et al. 2026 Environ Pollut*Calibration model for PurpleAir-monitored $\text{PM}_{2.5}$ across the United States*

Random-forest correction for PurpleAir $\text{PM}_{2.5}$ trained on 2019–2021 and evaluated on a held-out 2022, using only sensor-reported inputs — PM_{10} , $\text{PM}_{2.5}$, PM_{10} , temperature, pressure, relative humidity — with no external meteorology. Pearson r against co-located EPA monitors is $0.90\text{--}0.96$ across buffers from 500 m to 5 km; multiple calibrated products are released at different buffer radii. Deliberately interpretable rather than a deep network, and explicitly aimed at regions without regulatory infrastructure. *The limitation is the metric: r measures covariation, not agreement, and is blind to the multiplicative bias hygroscopic growth actually produces.* Without RMSE, slope or bias the accuracy claim is unverified. Still the most directly transferable record of the day for a distributed PM network — a national, sensor-only, temporally validated correction is the exact artefact a deployment needs.

doi: 10.1016/j.envpol.2026.128935

Benjamin et al. 2026 Sensors (Basel)*Limitations of environmental extrapolation in low-cost carbon monoxide and $\text{PM}_{2.5}$ sensors*

Multi-season co-location of a Plantower PMS5003 and an Alphasense CO-B4 against Federal Equivalence Method monitors, with calibration models deliberately trained on restricted ranges of relative humidity, temperature and pollutant concentration and then scored outside those ranges. **Within-envelope bias was minimal; crossing any envelope wall introduced bias; crossing the upper concentration wall specifically increased random error.** The design is the contribution — most co-location papers report a single in-sample R^2 and say nothing about the envelope. Limitations: one site, one sensor model per pollutant, and no aerosol-composition contrast, so the concentration effect cannot be separated from a possible composition shift at high concentration.

doi: 10.3390/s26154988

Bobbia et al. 2026 Data in Brief*Multi-source dataset of NO_2 and PM from mobile and fixed sensors in Rouen (France)*

BACKFILL (PubMed entry 2026-07-09; surfaced by the Amass connector, not by the date window). Data descriptor for the TIGA “Smart Mobility for All” deployment in Rouen: **10 mobile AtmoTrack** units on buses at 10 s, **18 fixed AirSenseEUR** units on traffic lights at 1 min, **6 reference stations** at 15 min, plus 2 AtmoTrack units co-located beside a reference station, over **15 January–30 November 2025**. Mobile records carry bus speed, GPS position and onboard RH/T/pressure; fixed records add O_x , NO and CO. The stated purpose is explicitly **multi-hop calibration** — transferring a reference-derived correction through fixed sensors to mobile units that never see a reference site — and heterogeneous data fusion. This is the single most directly usable record of the issue for distributed low-cost PM sensing, because it is the rare case where the co-location topology needed to test calibration transfer is published rather than described. Limitation: it is a data descriptor, so it contains no calibration result, no error metric and no validation; the value is entirely in the availability of the raw structure.

doi: 10.1016/j.dib.2026.113025

Cao et al. 2026 Atmos Environ*Characterising on-road NO_2 and particulate matter across urban road environments using city-scale mobile monitoring*

Abstract not deposited at Crossref on the entry date. The most directly relevant record of the day for mobile and distributed sensing and the one whose loss to the abstract famine costs most. Queued for the Saturday sweep.

doi: 10.1016/j.atmosenv.2026.122302

Cho et al. 2026 Sci Total Environ*Greenspace, land use and $\text{PM}_{2.5}$ across pandemic phases: a community scientist-led mobile sensing campaign*

Community scientists ran low-cost mobile $\text{PM}_{2.5}$ monitoring across Norwood, Ohio, for two summer periods — the initial pandemic phase (2020) and the reopening phase (2021) — with a meteorology-adjusted difference-in-differences framework for the year contrast and street-level linear land-use regression for spatial structure. **Higher 2021 concentrations were not fully explained by meteorology.** In the 2021 afternoon model, $\text{PM}_{2.5}$ was elevated within 50 m of office and manufacturing land use and inversely associated with NDVI within 100 m. **Cross-validated R^2 0.17–0.25 (2020) rising to 0.55–0.61 (2021).** The paper attributes the gap to source recovery, but a low-contrast summer is an equally good explanation and is not tested. Health-survey scores by ward are reported alongside but are not linked to measured exposure at the individual level. The transferable part is the QC and sampling protocol for neighbourhoods with no regulatory coverage, which is stated explicitly as the deliverable.

doi: 10.1016/j.scitotenv.2026.182124

Fawaz et al. 2026 Sensors (Basel)*Surface cleaning of SAW-based microparticle sensors in a cascade impactor via SAW-induced droplet displacement*

Rayleigh waves generated on a lithium-niobate chip actuate a water droplet that sweeps deposited particles out of the sensing zone of surface-acoustic-wave microparticle sensors integrated into a cascade impactor. Optimal droplet size and required RF power were determined; the droplet displaced both NaCl-solution residue and SiC particles without degrading sensor performance. Surface saturation is the binding constraint on long unattended deployment of mass-sensitive acoustic sensors, so a self-regenerating cycle changes the maintenance model rather than the sensitivity. Laboratory demonstration only — no ambient aerosol, no duty-cycle endurance data, and no accounting for what the droplet does to semi-volatile mass.

doi: 10.3390/s26154666

Gao et al. 2026 Atmos Meas Tech*Feasibility of an air sensor array for real-time detection and characterisation of VOCs*

Commercial VOC air sensors are broadband — each returns one scalar reflecting an ill-defined sensitivity envelope, which is why they have seen little research use. This paper tests whether an *array* spanning three fundamentally different sensor types can recover speciation from the joint response. The measurand is VOCs, not PM, and it is carried here for the method: the same argument applies directly to optical PM sensors whose single scalar folds together size distribution, refractive index and hygroscopic growth. Feasibility study — laboratory characterisation, and the practical question of whether the array's discrimination survives ambient interferent mixtures is not settled.

doi: 10.5194/amt-19-5337-2026

Gilardoni et al. 2026 Atmos Chem Phys*First continuous aerosol measurements at Testa Grigia at 3480 m a.s.l.: aerosol populations and dust transport dynamics in the southern European Alps*

First continuous **number size distribution and absorption coefficient** record at 1 h resolution from the Testa Grigia Observatory (3480 m a.s.l., Italian Alps), spanning **September 2021–May 2023**. Three aerosol population types are separated by transport pathway: a coarse population indicating long-range Saharan transport, a fine population tied to Po Valley and alpine-valley boundary-layer dynamics and broader continental flow, and low-number periods representing free-tropospheric background. The record is a baseline dataset rather than a hypothesis test, and the limitation is the one every single-site campaign carries — population attribution rests on back-trajectory and covariance reasoning, not on independent source measurement. Relevant to distributed sensing for a narrow but useful reason: a high-altitude site with characterised free-tropospheric periods gives an absolute-zero reference for optical instruments that low-altitude urban co-location cannot provide, and the absorption-coefficient record is one of the few instrument-side quantities published at deployment cadence.

doi: 10.5194/acp-26-11135-2026

Atmos Environ 2026b METADATA ONLY Atmos Environ*Stable carbon isotopic composition of PM₁ emissions from biomass and coal combustion under monitored combustion conditions*

METADATA ONLY — no abstract retrievable at build time. Carried because $\delta^{13}\text{C}$ signatures for PM₁ measured under *monitored* combustion conditions are the reference values that source-apportionment studies need and usually borrow from decades-old literature. Biomass and coal are the two sources whose separation matters most in residential-heating regions, and combustion-condition control addresses the standing objection that isotopic signatures drift with burn phase. If the reported ranges are tight this is a reference dataset; if they overlap it undermines $\delta^{13}\text{C}$ apportionment. Unresolvable without the text.

doi: 10.1016/j.j.atmosenv.2026.122296

Aerosol Air Qual Res 2026 METADATA ONLY Aerosol Air Qual Res*Identifying winter PM_{2.5} sources in an urban environment using receptor modeling and nonparametric trajectory analysis*

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar. Carried because the pairing of receptor modelling with *nonparametric* trajectory analysis is methodologically specific: it drops the functional-form assumption that conventional trajectory statistics impose on the source–receptor relationship, which is the assumption most likely to fail in complex winter meteorology with shallow boundary layers. Site, sample size and factor count are all unstated. Listed for the register, not weighted.

doi: 10.1007/s44408-026-00157-8

Kafi et al. 2026 Sci Total Environ*Evaluating the suitability of Sentinel-5P data for air quality monitoring in Tehran*

Six years (2019–2024) of TROPOMI Level-2 CO, NO₂, SO₂, O₃ and UV Aerosol Index against 22 ground stations in a megacity with complex topography, assessed by Pearson and Spearman correlation and Kling–Gupta Efficiency, plus a spatiotemporal trajectory comparison. **Only NO₂ showed strong agreement; CO, SO₂ and O₃ were weak**, and the trajectory analysis diverged as well. The authors are explicit that UVAI represents column aerosol loading rather than surface particulate concentration and warn against reading it as PM — a caveat routinely ignored in the satellite-PM literature and worth citing for that reason alone. The diagnosis (coarse pixel size against a topographically constrained urban basin) is plausible but not separated from the competing explanation of station siting, and no vertical-profile or boundary-layer-height information is brought in to test it.

doi: 10.1016/j.scitotenv.2026.182069

Kirchhoff et al. 2026 Atmos Meas Tech*Evaluation of DMSO as working fluid in condensation particle counters*

Laboratory and field comparison of a DMSO-operated CPC against a butanol counterpart across pressure, temperature and aerosol type, with a modified automatic refill system run for six months. Growth depends strongly on saturator temperature and saturator–condenser ΔT , confirmed by downstream optical particle counting and saturation/droplet-size simulation. **Mean cutoff 5.8±0.9 nm (DMSO) vs. 5.6±0.5 nm (butanol)**, with substantially lower fluid consumption; DMSO–H₂O mixtures extend the operating range. The wider cutoff spread on the DMSO unit is the honest cost, and matters if you are differencing two size channels. Non-flammability is what makes airborne and unattended remote deployment tractable.

doi: 10.5194/amt-19-5243-2026

Laffolley et al. 2026 ACS Omega*A 3D-printed two-stage virtual impactor (μ SPLIT) for radioactive aerosol size classification and direct analysis*

Stereolithography-printed virtual impactor separating aerosol into >10 , $1\text{--}10$ and $<1\ \mu\text{m}$ aerodynamic classes onto integrated filters for direct post-collection α counting, designed for Fukushima Daiichi decommissioning where the instrument must be cheap enough to incinerate rather than decontaminate. Flow path optimised by CFD with Lagrangian particle tracking; predicted cutoffs **9.0** and **1.3 μm** . The paper is unusually candid about its own failure modes: simulations identify small-particle contamination in the second-stage minor flow and large-particle contamination in the major flow, and dimensional metrology on the printed parts found channel contraction and corner rounding — i.e. the as-built geometry is not the as-designed geometry, which for an impactor moves the cutoff directly. Experimental validation is a proof of concept only: incense smoke, and Rn-progeny-bearing NaCl at $1.70\ \mu\text{m}$ GMAD putting peak α activity on the middle filter. No standardised-aerosol calibration and no quantitative wall-loss budget, both of which the authors name as required next steps. Relevant well beyond radiological work — additive manufacture of size-selective inlets is the cheapest available route to giving a low-cost node a real aerodynamic size cut instead of an optical inference.

doi: 10.1021/acsomega.6c04521

Li et al. 2026 Photoacoustics*Spatiotemporally synchronised dual-wavelength photoacoustic-scattering measurement of aerosol optical properties*

A sphere-tube coupled resonator that removes the transport, response-time and microenvironment mismatch inherent to split absorption/scattering instruments. Two integrating spheres act simultaneously as scattered-light collection cavities and as acoustic buffers forming a low-frequency **differential Helmholtz** resonator, while the central connecting tube supports a high-frequency **second-order longitudinal** mode; 450 and 532 nm lasers are modulated at **2572** and **169 Hz** and demodulated by an FPGA digital lock-in, so both wavelengths and both quantities are recovered from one sampling path. Absorption channels calibrated against NO_2 , scattering against N_2/CO_2 Rayleigh scattering; at 1 s integration and 3σ the 532 nm limits of detection are **$1.2\ \text{Mm}^{-1}$** (absorption) and **$1.32\ \text{Mm}^{-1}$** (scattering). The validation is the weak part: water mist, controlled combustion and a field check establish channel isolation and feasibility but not accuracy against a reference absorption method, and no Allan-variance drift figure is given, so the 1 s LOD does not tell you the useful averaging time. For distributed sensing the transferable idea is architectural rather than economic — this is a laboratory instrument, but co-locating absorption and scattering in a shared volume is what would let a node report single-scattering albedo instead of a mass proxy.

doi: 10.1016/j.pacs.2026.100864

Liu et al. 2026c Atmos Meas Tech*Polarization calibration of spaceborne lidar using dense cirrus-scattered solar background*

Solar background scattered by optically thick ice cloud provides a semi-continuous daytime polarization calibration that complements onboard pseudo-depolarizers. At 1064 nm (CATS) molecular scattering is negligible; at shorter wavelengths it polarizes the background and biases the calibration. The paper adds a vector-radiative-transfer molecular-scattering correction and shows the daytime bias at CALIOP 532 nm is $\sim 1\%$, removed by the correction to agree with onboard calibration; EarthCARE ATLID at 355 nm is treated by model calculation only. Carried at tier C: depolarization ratio drives satellite aerosol typing, which in turn feeds AOD-to- $\text{PM}_{2.5}$ conversion, but the link to surface concentration is two inferential steps away.

doi: 10.5194/amt-19-5325-2026

Liu et al. 2026d Sensors (Basel)*LRM-YOLO: lightweight YOLOv10n model for forest fire smoke detection in UAV images*

YOLOv10n with a RepViTBlock feature-extraction module, a lightweight efficient convolutional detection head, and MPDIoU bounding-box regression loss. Against baseline: **-36.7% parameters**, **-42.3% GFLOPs**, **$+1.3\%$ mAP50**, **$+3.7\%$ mAP50-95**, recall $+3.2\%$, precision $+3.6\%$, on a purpose-built UAV-perspective dataset plus Wildfire Smoke V1. Smoke detection in imagery is a plume-presence proxy, not a concentration measurement, and the self-constructed dataset means the headline gains are partly a property of the benchmark. Relevant only as an early-warning trigger upstream of a real exposure measurement.

doi: 10.3390/s26154887

Pérez-Pastor et al. 2026 Environ Sci Pollut Res Int*Robust quantification of anhydrosugars and reference material evaluation*

Accelerated solvent extraction (dichloromethane:methanol), BSTFA derivatisation, GC/MS for levoglucosan, galactosan and mannosan. Linearity $R > 0.9987$, repeatability $<6\%$, LOD $<0.02\ \text{mg L}^{-1}$; relative error $1\text{--}7.4\%$ against SRM 2787 (fine PM). **SRM 1649b (urban dust) gave extraction-condition-dependent levoglucosan values**, which the authors attribute to matrix ageing — a direct warning against using an old urban-dust SRM to validate a biomass-burning tracer method. Chromatographic calibration dominates the uncertainty budget. Expanded uncertainties meet forthcoming EU air quality monitoring requirements.

doi: 10.1007/s11356-026-38136-6

Shirikhani et al. 2026 J Aerosol Sci*Design and development of a high-throughput multipattern aerosol deposition device (MARTHA) for respiratory toxicity and health-risk assessment*

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar at the time of this issue (S2 returns 404 on the DOI). Listed rather than summarised. It is flagged as the day's highest-priority re-check for a specific reason: the title states an aerosol-delivery instrument built for respiratory toxicity assessment, which is the exact instrument $\times$ health-endpoint coupling that the 2–8 August weekly identified as structurally absent across 146 records. Nothing beyond the title and the author list (Shirikhani, Salana, Amanatidis, Washeem, Eiguren-Fernandez, Spielman, Nannu Shankar, Lewis) supports that reading yet.

doi: 10.1016/j.jaerosci.2026.106882

Wang et al. 2026 Atmos Environ*Measuring heavy metals in atmospheric particulate matter: operational definitions, method comparability and environmental interpretation*

METADATA ONLY — no abstract in Crossref, Europe PMC or Scholar Gateway. Listed, not summarised. Carried because the title states the problem this watch treats as central, transposed to a different measurand: for heavy metals in PM, the reported concentration depends on the digestion protocol, the filter substrate, the size cut and whether total or bioaccessible fraction is meant — so “comparability” is a property of the method chain, not of the air. The same argument applies to low-cost optical mass measurement, where the reported $\mu\text{g/m}^3$ is a function of assumed density, refractive index and hygroscopic growth. A systematic treatment of operational definitions on the metals side would be worth reading against the sensor-calibration literature.

doi: 10.1016/j.atmosenv.2026.122291

Wu & Qu 2026 Environ Monit Assess*Interpretable short-term PM_{2.5} forecasting using meteorology–pollution coupling across Beijing stations*

420,768 hourly observations from 12 Beijing stations, 2013–2017, under a strict chronological train–validation–test split with Diebold–Mariano testing, feature-group ablation and station-holdout transfer. At the 6 h horizon CatBoost gave **RMSE 51.24 $\mu\text{g}/\text{m}^3$, MAE 30.87**, $R^2 = 0.617$, a 6.73% RMSE reduction over persistence; station-holdout LightGBM reached 53.11 $\mu\text{g}/\text{m}^3$, 5.65% below persistence. Dominant predictors were recent PM history, wind speed and direction, diurnal cyclicity and the temperature–dew-point gap. **The honest part is the framing:** a 6.7% gain over persistence is a modest operational return, performance degrades during high-pollution episodes, and the authors state that the 2013–2017 regime would require retraining for contemporary deployment.

doi: 10.1007/s10661-026-15799-9

Yaqoob et al. 2026 Sci Rep*Semiparametric hybrid air quality forecasting using Prophet ensemble and feature selection*

Prophet models the trend and seasonality of daily PM_{2.5}; the residuals are then modelled by bagging, boosting, stacking and voting regressors, with Lasso, SHAP and recursive feature elimination applied to control the high-dimensional lag/interaction feature space. Tested on Seoul daily data and a high-dimensional simulated dataset against XGBoost, SVR and LSTM baselines; Prophet + boosting with Lasso and SHAP gave the lowest RMSE. The residual-modelling decomposition is a sensible way to keep an interpretable trend component, but the abstract reports no RMSE values, no forecast horizon and no comparison against a persistence baseline, which is the benchmark that actually discriminates in daily PM forecasting.

doi: 10.1038/s41598-026-64590-w

Zhang et al. 2026 Atmos Chem Phys*High frequency of urban new particle formation on the Tibetan Plateau*

In-situ particle number size distribution measurements at Tibet University, Lhasa (3,650 m), across spring and autumn campaigns, with days classified NPF / Undefined / Non-Event from PNSD evolution plus size-segregated counts and growth rates. NPF event frequency was **86.2% in autumn and 62.1% in spring** — extremely high by urban standards — and estimated CCN was $(2.3 \pm 1.5) \times 10^3 \text{ cm}^{-3}$ on NPF days versus $(1.3 \pm 0.7) \times 10^3$ on non-event days. Autumn events were associated with greater sulfur-precursor availability; no dominant control was identifiable in spring, which the authors state plainly. Relevant here as a reminder that number-based instrumentation resolves a regime mass-based monitoring cannot see at all; the high-altitude, low-condensation-sink setting limits transfer to lowland urban networks.

doi: 10.5194/acp-26-11473-2026

Zhang et al. 2026b Atmos Meas Tech*Impact of aerosol-type assumption on Landsat 8 atmospheric correction over land*

600 Landsat 8 scenes over 100 AERONET sites, comparing the Urban Clean aerosol type hardwired into the LaSRC surface-reflectance algorithm against MOD04- and MOD09-based dynamic types. **MOD04 gave the best AOD accuracy:** $R^2 = 0.7236$, **RMSE 0.0437**, **bias 0.0052**. Relevant here not for surface reflectance but because the same fixed-aerosol-type assumption propagates into satellite AOD products used as predictors in satellite–surface PM_{2.5} fusion models — an R^2 of 0.72 against AERONET is the ceiling any such predictor inherits. AERONET sites are systematically non-representative of high-PM urban environments, so this is likely optimistic.

doi: 10.5194/amt-19-5281-2026

Zhang et al. 2026c Atmos Environ*Characterisation of air pollution transport using absorbing aerosol index from multiple payloads*

Abstract not deposited at Crossref on the entry date. Multi-payload AAI synergy is a transport-attribution method rather than a surface-concentration one; logged for completeness. Queued for the Saturday sweep.

doi: 10.1016/j.atmosenv.2026.122298

Atmos Environ 2026a METADATA ONLY Atmos Environ*Reliability-constrained identification of primary and elevated aerosol structures from a single tilted Mie lidar under dust-influenced conditions*

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar. Listed, not summarised. Carried because the title states a specific and testable capability: retrieving both the primary and the elevated aerosol layer from a *single tilted* Mie lidar, with an explicit reliability constraint, under dust loading that normally defeats single-wavelength inversion. Single-instrument retrieval of layer structure is the difference between a research lidar and one a network can afford, and dust is the condition under which the retrieval usually fails. Whether the paper delivers that is unknown until an abstract appears.

doi: 10.1016/j.atmosenv.2026.122297

d'Hondt et al. 2026 Research Square (preprint)*Neural downscaling of air-quality simulations requires structural correction before spatial refinement*

Deep-learning downscaling of CAMS-class chemical transport model output, which is capped near 10 km for continental forecasts. The argument is that existing approaches either treat the coarse field as a smoothed copy of the fine field or learn a direct coarse-to-fine map, and both conflate two different things: cross-resolution differences in *model behaviour*, and genuinely missing spatial detail. Separating a structural correction step from the super-resolution step is what the paper proposes and tests. Directly transferable to LCS-plus-CTM fusion, where the network supplies the detail the model lacks and the model supplies the physics the network lacks — but a bias-corrected coarse field is the precondition. Preprint, not peer reviewed; carried on relevance, and no numeric claim from it is used in this issue's analytics.

doi: 10.21203/rs.3.rs-10627504/v1

Exposure assessment & modelling**26 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (26 OF 26 RECORDS IN THIS SUBTOPIC)

Adeniji et al. 2026 PLOS Glob Public Health*Long-term air quality and meteorological coupling dynamics in Abeokuta, Nigeria*

Twelve years of hourly PM_{2.5}, PM₁₀, NO₂, SO₂, CO and O₃ (2012–2023) against six meteorological variables at a single West African site. Particulate matter dominates and **consistently exceeds WHO guideline values**, peaking in the December–March Harmattan; regression against temperature gives R^2 **0.48–0.52**, with negative associations for rainfall and humidity consistent with thermal stagnation and wet scavenging. The aggregate AQI stays in the “moderate” band (39–51) while the health-risk-adjusted AQI exceeds **90** seasonally — the paper’s real contribution, since it demonstrates that an index which averages across pollutants conceals a PM-driven risk that a health-weighted index exposes. Two load-bearing limitations: the data provenance for a 12-year hourly record in Abeokuta is not established in the abstract, and reanalysis or satellite-derived series are common substitutes at such sites; and single-site, single-city inference cannot separate the meteorological coupling from local source seasonality (biomass burning peaks in the same months as the Harmattan). Sub-Saharan Africa remains the most under-monitored region in this watch and this is the first record from it in six issues.

doi: 10.1371/journal.pgph.0006683

Chebaicheb et al. 2026 Environ Int*A phenomenology of submicron organic aerosol components in urban France*

The first nationwide harmonised organic-aerosol source apportionment for France: time-resolved Aerosol Chemical Speciation Monitor measurements at 12 (sub)urban sites over one to seven years each, with a single rolling PMF protocol applied under common constraints and acceptance criteria across all datasets. Primary factors — hydrocarbon-like, biomass-burning and cooking-like OA — account for **≈33% of OA mass**; oxygenated OA dominates at every site and splits into less- and more-oxidised components everywhere except Poitiers and Strasbourg. Site-specific factors resolved a mixed shipping/industrial component at Marseille Longchamp and an amine-related component at Strasbourg and Creil. Wintertime LO-OOA correlates with BBOA, implicating residential combustion in secondary as well as primary mass; summer LO-OOA tracks biogenic precursors. **CHIMERE shows systematic component-level bias against the observations**. The harmonisation is the contribution — rolling PMF applied inconsistently is why the existing European OA literature does not pool — and the limitation is that (sub)urban sites only cannot constrain the rural background these factors are transported through.

doi: 10.1016/j.envint.2026.110453

Chen et al. 2026 Environ Sci Technol*Inventory uncertainty and transboundary transport of biomass-burning black carbon in southern China*

Year-round radiocarbon measurement of biomass-burning black carbon at two regional and one background site in southern China, compared against tailored atmospheric-transport simulations. The model reproduces seasonality but the normalised mean bias spans **–68% to +28%** — a factor-of-two-plus spread in a quantity inventories report without error bars, attributed largely to unmonitored biomass burning in mainland China. Residential burning across East Asia dominates; Southeast Asian transport matters in summer and, via free-tropospheric pathways, **persists in winter** when continental air masses are assumed to exclude it. *Limitation*: three sites cannot constrain a regional inventory, and ¹⁴C apportionment inherits uncertainty from the assumed fossil/non-fossil end-members. Relevance: this is the cleanest current demonstration that source-diagnostic *measurement*, not finer model resolution, is what bounds inventory error — the argument for putting source-resolving capability into distributed networks.

doi: 10.1021/acs.est.6c00449

Chillrud et al. 2026 Environ Epidemiol*pcpr: principal component pursuit for exposure pattern recognition and outlier detection*

R package implementing PCP with three environmental-health extensions: a non-negativity constraint on the low-rank matrix, missingness handling, and a specialised penalty for below-LOD observations. Demonstrated on 26 PM_{2.5} constituents measured every few days in Queens, NYC, 2015–2021: four patterns recovered (secondary/tailpipe, traffic, crustal dust, salt) plus **211 outlying events** including July Fourth fireworks spikes. PCP’s sparse component is an outlier detector, not a source; calling the 211 events “exposure events” presumes an interpretation the decomposition does not supply. Still the most immediately usable artefact in today’s corpus for anyone holding speciated network data.

doi: 10.1097/EE9.0000000000000515

Deng et al. 2026 Atmos Chem Phys*Impact of black carbon on daytime valley and slope winds in idealised simulations*

High-resolution idealised WRF-Chem experiments, run with and without radiatively interactive black carbon, on thermally driven valley circulation. Absorption warms the upper boundary layer and cools lower levels, increasing stability, reducing surface heating, and leaving up-slope winds **weaker and confined to a shallower layer**. The feedback is self-reinforcing: the circulation that ventilates a valley is suppressed by the aerosol it would otherwise disperse. *Limitation*: two idealised simulations with prescribed BC, no observational constraint, no real terrain — the sign of the effect is credible, the magnitude is not transferable. Relevance to monitoring: in mountain basins the relationship between emissions and measured concentration is not linear, and siting a network to sample a circulation that the pollution itself modifies requires the vertical dimension, not a denser surface grid.

doi: 10.5194/acp-26-11171-2026

Environ Sci Atmos 2026a Environ Sci Atmos*Multi-scale variability and lagged meteorological controls of PM_{2.5} in Delhi*

Wavelet coherence and causality analysis of Delhi PM_{2.5} against meteorological drivers, decomposing the record by timescale rather than fitting a single regression. **The abstract is truncated at source** — it stops mid-sentence after stating the aim, and no coherence bands, lag lengths, data period or station count are recoverable. What can be said is that the method is the right one for the problem: Delhi’s PM_{2.5} is driven by processes on separable scales (diurnal boundary-layer collapse, synoptic ventilation, seasonal crop-residue burning) that a time-domain model conflates, and lagged causality is what forecasting a distributed network needs. Carried for the register with its numbers unstated. Re-check on the weekly abstract sweep.

doi: 10.1039/d6ea00071a

Fan et al. 2026 Atmos Environ*Santa Ana winds control wildfire smoke transport and related health impacts in California during January 2025*

METADATA ONLY — no abstract in Crossref, Europe PMC or Scholar Gateway. Listed, not summarised. Carried because it is the first record in this watch to attach a health-impact estimate to the January 2025 Los Angeles fires, and because the physical premise is specific and testable: offshore Santa Ana flow reverses the usual coastal transport direction, so smoke-attributable exposure is not predicted by proximity to the burn perimeter. Satellite work on the same event (Li et al., *AGU Advances*, Feb 2026) reports spread rates to **3.7 km h^{–1}**, residential-area fire intensities of **72–297 W m^{–2}** and a total-PM emission coefficient of **10.2–21.1 g MJ^{–1}**, with 24 h mean PM_{2.5} within 7.5 km reaching **300 µg/m³** — context, not this paper’s result. The health-impact methodology is unknown until an abstract appears, and for a wildland–urban-interface fire burning structures the applicability of vegetation-fire concentration–response functions is exactly the assumption that needs checking.

doi: 10.1016/j.atmosenv.2026.122292

Florêncio et al. 2026 Atmos Environ

Comprehensive analysis of PM₁₀ in the Metropolitan Area of São Paulo, Part I: chemical characterisation and sources

Abstract not deposited at Crossref on the entry date. Title indicates chemical speciation and source apportionment for a megacity with almost no component-resolved PM literature in this watch to date. Queued for the Saturday sweep.

doi: 10.1016/j.atmosenv.2026.122286

Gong et al. 2026 Res Sq (preprint)

Regional analysis of urban physical factors influencing air pollution: evidence from Seoul's living areas

Constructs a **100 m** grid over the Seoul metropolitan area, kriges a continuous surface from **193 monitoring stations** for SO₂, CO, NO₂, O₃, PM₁₀ and PM_{2.5} combined into a Comprehensive Air-quality Index, reduces 15 urban-form variables to four principal components (urban enclosure, low-density buildings, high-density buildings, development intensity) and fits geographically weighted regression. CAI rises from **88–92** in northeastern Seoul to **96–104** in the southwest. The headline model comparison — adjusted **R² 0.9928** for GWR against **0.0182** for OLS — should be read as a warning rather than a result. A 54-fold jump in explained variance on a kriged surface means the GWR is largely reproducing the interpolation, since kriged values at 100 m between 193 stations are smooth by construction and locally varying coefficients will fit that smoothness almost exactly. No spatial cross-validation or effective-degrees-of-freedom correction is reported. Tier C. Carried because the urban-form-to-exposure framing is the right one for siting a sensor network, and because it is a clean illustration of how interpolated exposure surfaces manufacture model fit.

doi: 10.21203/rs.3.rs-10443285/v1

Gould & Jagarnath 2026 Research Square (preprint)

Satellite monitoring for air quality and public health research in Africa: a scoping review

JB1 methodology, PRISMA-ScR reporting, five databases plus grey literature to May 2026, for studies using satellite-derived air quality data against human health outcomes in Africa. **Twenty-six records met inclusion** — for an entire continent. Most post-2015, most ecological time-series or cross-sectional, concentrated in West Africa, with PM_{2.5} the dominant exposure. This is the first record in this watch to put a number on the African evidence gap that its own geography figure has shown as an empty bar every day since 27 July. Preprint, not peer reviewed; a scoping review counts literature and does not weigh it.

doi: 10.21203/rs.3.rs-10623736/v1

Atmos Environ 2026c METADATA ONLY Atmos Environ

Source-specific black carbon responses to short-term interventions and cross-year emission changes in coastal Qingdao

METADATA ONLY — no abstract retrievable. Carried because source-specific BC response to *short-term* interventions is a natural-experiment design, and coastal sites separate shipping from land-based combustion in a way inland sites cannot. Read alongside Chen et al. in this issue, which finds inventory-based BC attribution biased by –68% to +28%: an intervention study measures the response directly and sidesteps that. Sample period, apportionment method and intervention definition are all unstated.

doi: 10.1016/j.atmosenv.2026.122299

Hernandez et al. 2026 Atmos Environ

Urbanization and humidity impact atmospheric deposition of anthropogenic microparticles (including microplastics) in a humid subtropical city

METADATA ONLY — no abstract in Crossref, Europe PMC or Scholar Gateway. Listed, not summarised. Carried at tier C because atmospheric microplastic is a particle class that optical monitors detect but do not identify: fibres in particular violate the spherical-equivalent-diameter assumption underlying every optical particle counter, so a deposition record that quantifies their abundance as a function of urbanisation and humidity bears on the size and shape distribution a low-cost sensor is actually sampling. The humidity dependence is the part worth reading if an abstract appears, since RH is simultaneously the dominant artefact in optical PM measurement and, here, an exposure variable.

doi: 10.1016/j.atmosenv.2026.122294

Ibrahim 2026 Research Square (preprint)

Air pollution status and temporal patterns in Bukit Rambai and Muar, 2013–2017

Five years of Malaysian Department of Environment station data (API, O₃, CO, SO₂, PM₁₀, NO₂) at two southern Peninsular stations, analysed in openair with calendar plots. Peak API 477 (Bukit Rambai) and **762 (Muar)** in June 2013 during a transboundary haze episode. Descriptive, two stations, no exposure or health linkage, and not peer reviewed. Carried only because episodic Southeast Asian haze is otherwise unrepresented in this watch and the 762 value is a useful anchor for dynamic-range requirements on low-cost optical sensors, which saturate well below it.

doi: 10.21203/rs.3.rs-10666765/v1

Li et al. 2026a Environ Sci Technol

Upgrading diesel vehicle after-treatment technology significantly changed the composition of reactive nitrogen in exhaust

Chassis-dynamometer and controlled testing of light-duty diesel vehicles across China IV, V and VI standards. Nitrogen speciation shifts fundamentally: **NO exceeds 80% of reactive nitrogen under China IV**, while NH₃ and N₂O become prominent under China VI. Mechanisms are attributed component-wise — diesel oxidation catalysts raise the NO₂/NO_x ratio, SCR produces NH₃ slip, ammonia-slip catalysts generate N₂O. SHAP analysis of tree models identifies emission standard, exhaust temperature and vehicle speed as dominant drivers. *Limitation*: no fleet sample size or vehicle count is given, and dynamometer cycles under-represent cold starts and real-world transients where SCR is least effective. Relevance: NH₃ is a secondary-PM precursor and N₂O a greenhouse gas, so successive NO_x controls have moved the problem rather than closed it. A traffic-adjacent network measuring NO_x alone now under-reports the emission it is meant to characterise.

doi: 10.1021/acs.est.6c00994

Lin et al. 2026 Atmos Chem Phys

Elevated aerosol layers within the daytime mixed layer over ecologically sensitive areas of northwest China: diurnal variation, formation mechanisms and regional effects

Identifies an elevated aerosol layer embedded *within* the daytime mixed layer over the Hexi Corridor using intensive vertical-profile campaigns in **2010 and 2012** at Dunhuang and Minqin. Excluding dust events, elevated layers occurred at **0.6–2 km** on **82 %** of days at Dunhuang and **44 %** at Minqin, with the paper analysing formation mechanism, air-quality consequence and radiative effect. The relevance to surface monitoring is direct and under-appreciated: a concentration maximum aloft but inside the mixing layer breaks the well-mixed assumption that lets a surface monitor stand for a column, and it is precisely the configuration in which satellite AOD and ground PM_{2.5} decouple — which is the failure mode of every AOD-to-surface-PM scaling relationship. Limitations: two campaigns fourteen years old, two sites, and an arid region whose boundary-layer behaviour does not generalise to the humid urban settings where most low-cost networks are deployed.

doi: 10.5194/acp-26-11111-2026

Lin et al. 2026 Environ Sci Technol

Three-millennia alpine-lake records of divergent atmospheric Cd, Pb, Sb and Hg pollution in southwestern China

Radiometrically dated sediment cores from three alpine lakes at ~4000 m in Yunnan reconstruct 3000 years of atmospheric Sb, Pb, Cd and Hg deposition. Onset is metallurgy-specific and spatially staggered: Sb and Pb from the mid-1400s in northeastern Yunnan, ~200 years later in the west with Cu/Ag smelting; Cd only after the early 1700s with Zn and Cu production. Preindustrial peaks fall between the early 1800s and 1900 CE. **Since the 1950s Cd has reached unprecedented levels while Sb and Pb remain below their preindustrial maxima**, and Hg rises monotonically from the mid-1400s. *Limitation*: lake sediment integrates catchment erosion as well as atmospheric deposition, and the separation depends on normalisation assumptions not visible in the abstract. Relevance: a 3000-year baseline is the only way to know what “background” means for a metal, and metal-resolved PM is where low-cost sensing has no capability at all.

doi: 10.1021/acs.est.5c17960

Makacha et al. 2026 Front Glob Womens Health

Spatio-temporal and structural determinants of personal PM_{2.5} exposure in three sub-Saharan African geographies

Five days per season of minute-resolved personal PM_{2.5} monitoring on 343 parous women in The Gambia, Kenya and Mozambique, 3,190 person-days, aligned to microenvironment, fuel use, occupation, wealth and meteorology, with facility-based ambient monitors for comparison. Daily medians were modest — 10.4 (7.6–14.7), 8.1 (6.8–9.8) and 10.9 (9.3–13.5) $\mu\text{g}/\text{m}^3$ — but episodic spikes dominated, and WHO 24-h exceedance was strongly seasonal (The Gambia **55.9% dry vs 13.6% wet**). **Ambient-personal concordance was 0.01–0.56 with slopes <1 throughout**. Exposure was higher away from home with a 12:00–16:00 peak, so clean-fuel transition alone will not close the gap. The 5-day-per-season design is thin for characterising episodic exposure and the authors do not report how much of the between-person variance it captures.

doi: 10.3389/fgwh.2026.1843221

Park et al. 2026 Environ Pollut

City-scale non-exhaust PM₁₀ from emissions to exposure with a coupled chemistry-CFD framework

Two non-exhaust emission-factor frameworks (EMEP/EEA Guidebook and KIMM) implemented in the Fine-scale Air Quality Simulation Unit, a 3-D coupled CFD-chemistry model, over a densely built Seoul district, combined with the 2022 Seoul Time-Activity Profile. **Total non-exhaust PM₁₀ was similar under both frameworks; source partitioning was not** — brake wear dominant under EMEP/EEA, road wear under KIMM. Non-exhaust reached **up to 70% of total PM₁₀** in near-road and dense cells; outdoor activity-weighted contributions of 9–10% in commercial and traffic zones exceeded concentration-based estimates because peak traffic coincides with peak outdoor presence. No measurement validation of the partitioning is presented, which is the obvious next step and the reason a receptor-model or tracer campaign would be complementary.

doi: 10.1016/j.envpol.2026.128941

Ray et al. 2026 Atmos Environ

Evaluating CCN activity through size-resolved hygroscopicity: synergistic observations and simulations in a tropical montane region

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar. Listed, not summarised. Size-resolved hygroscopicity (κ) is the physical quantity underneath every relative-humidity correction applied to an optical PM sensor, and tropical montane conditions are the high-RH regime where those corrections fail worst, so this is a priority re-check even though the paper’s own framing is cloud-condensation nuclei rather than air quality.

doi: 10.1016/j.atmosenv.2026.122284

Stevens et al. 2026 Atmos Environ

Identifying unique biomass burning marker fingerprints for Australian hardwoods, softwoods and grasses

Abstract not deposited at Crossref on the entry date. Fuel-specific tracer fingerprints are the missing input for wildfire source attribution outside the Northern Hemisphere; pairs with Pérez-Pastor et al. below on the metrology side. Queued for the Saturday sweep.

doi: 10.1016/j.atmosenv.2026.122303

Sui et al. 2026 J Environ Manage

Contrasting annual variation of organic and elemental carbon in the Pearl River Delta

Three years of continuous OC/EC observation (2019–2021) spanning the COVID-19 lockdown, with PMF source apportionment. OC declined steadily, driven by suppressed secondary organic carbon formation in the PRD’s NO_x-rich regime — a relationship the authors report as *reversing* during lockdown when NO_x fell sharply, which is the interesting result and the one that identifies the regime. EC was flat or slightly rising, so fossil combustion from industry and mobile sources has not responded to the control programme. **Fossil-related emissions are ~70% of carbonaceous aerosol** and that share did not fall over the period. A single-site three-year record cannot separate emission change from meteorological variability, and no such adjustment is described; the lockdown period is doing most of the causal work and is a one-off natural experiment with a confounded meteorology.

doi: 10.1016/j.jenvman.2026.130708

Atmos Pollut Res 2026 METADATA ONLY Atmos Pollut Res

High-resolution CFD modelling of traffic-related air pollutant dispersion in an urban area of Istanbul

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar. Carried because building-resolving CFD is the reference against which cheap dispersion proxies — Gaussian plume, land-use regression, distance-to-road — are judged, and because street-canyon gradients at the scale CFD resolves are exactly what a dense low-cost network claims to capture and what a regulatory monitor cannot. Whether the model is validated against measurement, and at what resolution, is unknown. Listed, not weighted.

doi: 10.1016/j.apr.2026.103169

You et al. 2026 Environ Pollut

Satellite-based estimation of ambient ammonia and emission inventory uncertainty in South Korea

Monthly ground-level NH₃ at 15 km resolution for 2013–2017, from Cross-track Infrared Sounder retrievals refined by a two-stage framework: a linear mixed-effects model carrying meteorology plus the Clean Air Policy Support System inventory, then a generalised additive model over the LMM residuals. **Cross-validated $R^2 = 0.73$, MAE 0.15, RMSE 0.21**. Concentrations peak in agricultural areas March–June; temperature is the dominant meteorological control (**+18.9% per +1 °C in June**), with relative humidity and wind speed strongest in March. The GAM residual surface identifies agricultural hotspots the inventory does not contain, which is the paper’s actual claim — inventory gaps, not model skill. NH₃ matters here as the limiting precursor for ammonium nitrate and sulfate, so a 15 km monthly NH₃ field is a plausible predictor for secondary PM_{2.5} models, but the resolution is far coarser than the sources, and validation against sparse ground NH₃ is circular with the sparsity that motivated the work.

doi: 10.1016/j.envpol.2026.128903

Zhang et al. 2026 Environ Geochem Health*National-scale heavy metal contamination in urban road dust across 82 Chinese cities*

13,269 samples pooled from 169 studies into an urban-agglomeration framework; geo-accumulation index for anthropogenic enrichment, potential ecological risk index, multivariate source deconvolution. Cd and Hg are the priority pollutants, jointly >85% of total ecological risk in every agglomeration despite low absolute concentrations relative to Mn and Zn; a Cd/Hg I_{geo} ratio is proposed as a source fingerprint. Pooling 169 studies with heterogeneous sampling depth, particle-size cut and digestion protocol is the central weakness — the fingerprint may partly index methodology. Road dust is the resuspension reservoir behind non-exhaust PM, so this pairs with Park et al. above.

doi: 10.1007/s10653-026-03422-2

da Costa Cardoso et al. 2026 Environ Monit Assess*Legacy Pb–Zn mine tailings as persistent sources of bioaccessible toxic elements in urban dust*

18 urban dust samples and reference soils from Boquira, a former Pb–Zn district in northeastern Brazil; microwave-assisted digestion and XRF for Pb, Zn, Cd, Cu, Mn, Fe, Al; geoaccumulation index, contamination and enrichment factors, ecological risk index; oral bioaccessibility by dilute-nitric-acid gastric simulation. **Pb bioaccessibility ~75%**, Cu 54.8%, Mn 28.1%, Zn 27.0%. Oral ingestion dominated over dermal; non-carcinogenic HI exceeded 1 for children on Pb, and carcinogenic risk from Cd exceeded limits for both children and adults. Eighteen samples in one district, a single gastric extraction protocol without an intestinal phase, and no particle-size fractionation — which matters, because resuspension and inhalation-relevant dose are size-dependent and this design cannot resolve them.

doi: 10.1007/s10661-026-15743-x

de la Torre et al. 2026 J Hazard Mater*A 15-year atmospheric record of intentionally produced persistent organic pollutants in Spain*

Polyurethane-foam passive samplers at **23 sites, 2009–2023**, evaluating Stockholm Convention effectiveness. Hexachlorobenzene dominates at a median **41 %** of $\sum$ POPs with a remote-enriched spatial pattern; legacy pesticides and indicator PCBs show temperature-dependent secondary volatilisation from contaminated soils by Clausius–Clapeyron analysis, and an *urban* enrichment plausibly driven by heat-island effects; PBDEs show a **35-fold** urban-to-remote gradient from late-regulated BDE-209. Early-banned compounds decline up to **–33 %/yr** while PeCB, iPCBs and BDE-209 are **stalled or rising to +20 %**. This is semi-volatile organic chemistry, not particulate mass, and enters on method rather than substance: a passive sampler is the cheapest long-record instrument in atmospheric science and this is a clean demonstration of what a 15-year, 23-site, zero-power network buys — an altitudinal profile on Tenerife separating local secondary emission from free-troposphere clearance. The tradeoff is equally clear: no time resolution, no size information, and partitioning between gas and particle phases is unresolved throughout.

doi: 10.1016/j.jhazmat.2026.143155

Cardiovascular & metabolic**7 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (7 OF 7 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026 Research Square (preprint)*Compound cold wave–PM_{2.5} exposure and cardiovascular emergency ambulance calls*

82,714 cardiovascular emergency ambulance call records from Shijiazhuang, 1 January 2014 to 31 December 2023, with three cold-wave categories defined on temperature percentiles crossed with four PM_{2.5} pollution levels to construct compound exposure events; distributed lag non-linear model with quasi-Poisson regression for lagged and interactive effects. The compound-exposure framing is the contribution — most temperature $\times$ PM work tests a multiplicative interaction term rather than constructing joint exposure states. Preprint, not peer reviewed; ambulance calls are a severity-filtered and access-dependent outcome; and a 3 $\times$ 4 exposure grid over ten years leaves the extreme cells thin.

doi: 10.21203/rs.3.rs-10547139/v1

Gaggini & Vassalle 2026 Int J Mol Sci*Gene–air pollution interaction in cardiovascular disease*

Narrative PubMed review of combined genetic-susceptibility and air-pollution effects on cardiovascular risk, concentrating on detoxification and inflammation genes. Argues the area could refine cardiovascular risk prediction beyond traditional factors. Narrative, not systematic; gene–environment interaction in this literature is dominated by underpowered candidate-gene studies, which the review acknowledges as “lights and shadows” but does not quantify. Useful only as an entry point to the literature.

doi: 10.3390/ijms27156818

Kakhaia et al. 2026 Environ Int*Exposome approach to causal effects of time-varying air pollution mixtures on CVD incidence*

UK Biobank, annual PM_{2.5} and NO₂ from a Europe-wide LUR model, 10-year cumulative CVD incidence under modified treatment policies. Confounders selected by a causal structure discovery algorithm with prior knowledge; sequential doubly robust non-parametric estimation with ensemble learning and censoring adjustment. A 25% exposure reduction over 2005–2019 gave **–0.62 per 100** (95% CI **–0.70 to –0.55**), RR $\approx$ 0.90; the projected net-zero scenario reached **–1.05 per 100** and RR $\approx$ 0.85. **The stated sensitivity is the finding:** dropping assessment-centre region from the adjustment set more than doubled the estimate, so the headline number is contingent on a spatial-confounding choice the authors cannot adjudicate. Method-forward, estimate-fragile.

doi: 10.1016/j.envint.2026.110441

Moro et al. 2026 Environ Res*Long-term PM_{2.5}, hypertension and metabolic syndrome in rural Ghana*

Cross-sectional analysis of 1,833 AWI-Gen participants aged 40–60 in rural Ghana, with 2007–2015 mean PM_{2.5} from a satellite-based model and MetS by harmonised criteria, under sequentially adjusted multivariable logistic regression. **OR 1.06 (1.02–1.11) per 1 $\mu\text{g}/\text{m}^3$ for hypertension**, a monotone quartile gradient ($p_{\text{trend}} = 0.003$) with the top quartile at 50% greater odds, and **PAF 18.6%**. **No association with MetS overall or with any other component** — the null is as informative as the positive, and the authors report it rather than burying it. Three limitations bound the inference: the design is cross-sectional so temporality is not established; exposure is modelled at satellite resolution over a region with almost no reference monitoring and no local validation is reported; and household air pollution from solid-fuel cooking, which dominates personal exposure in this setting, is not in the model at all.

doi: 10.1016/j.envres.2026.125484

Ting 2026 JACC Adv*Statistical fragility and methodological constraints in metabolomic mediation of air pollution and cardiovascular risk*

Metadata-only: correspondence deposited without an abstract. Registered because it completes an exchange this watch is already carrying — the authors' reply (10.1016/j.jacadv.2026.103149) surfaced in the 12 August window as an Elsevier metadata-only record and remains unresolved. The subject, statistical fragility of mediation estimates in air-pollution metabolomics, is directly relevant: mediation analysis on high-dimensional metabolite panels with modest cohort sizes produces indirect effects whose confidence intervals are dominated by the product-of-coefficients variance, and the literature routinely reports them as though they were primary. No numeric claim is made from this record. Retrieve both letters together once full text is available.

doi: 10.1016/j.jacadv.2026.103150

Wittkopp et al. 2026 Am Heart J*Cleaner Air for Lower Cardiometabolic Risk: HEPA filtration trial protocol in prediabetes*

Protocol for a double-blind, randomised, sham-controlled trial of one month of bedroom HEPA filtration in adults with prediabetes in the New York City area (NCT05994937), with continuous indoor PM_{2.5} measured throughout. Primary outcomes are continuous glucose monitoring metrics before and after; exploratory outcomes include insulin resistance, serum biomarkers and transcriptomics. Secondary analyses will model measured PM_{2.5} as a continuous independent variable, which is the analysis that will actually be informative — the intention-to-treat contrast in filtration trials is routinely diluted by baseline indoor concentrations already being low. One month is far too short for any glycated-haemoglobin endpoint, which is presumably why continuous glucose monitoring was chosen instead. No results; logged to `trials.json`.

doi: 10.1016/j.ahj.2026.107573

Zhang et al. 2026 Zhonghua Yu Fang Yi Xue Za Zhi*Combined cold-spell and heavy-pollution exposure and circulatory mortality, Beijing–Tianjin–Hebei*

Daily mortality, PM_{2.5} and meteorology from 70 districts and counties, cold seasons (November–March) 2013–2018, mean daily PM_{2.5} **98.58 ± 78.19 μg/m³**. Three combined-exposure scenarios were constructed — overlapping cold spell and heavy pollution (A), a standalone pollution event within 7 d of a cold spell ending (B), and a standalone cold spell within 7 d of a pollution event ending (C) — with district-level DLNMs pooled by random-effects meta-analysis. The overlapping event gave **CRR 1.35 (1.14–1.60)** against **1.14 (1.09–1.20)** for cold spell alone and **1.08 (1.00–1.16)** for heavy pollution alone. Supra-additivity is the substantive result, and it is the case for joint rather than single-hazard alerting. Heavy pollution is defined by a concentration threshold, not by composition, so the interaction could equally be a composition effect: winter heating episodes in this region differ chemically from other high-mass days, and the design cannot separate the two.

doi: 10.3760/cma.j.cn112150-20260105-00006

Neuro / mental health**5 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (5 OF 5 RECORDS IN THIS SUBTOPIC)

Delaney et al. 2026 Environ Epidemiol*Cumulative annual air pollution exposure and first hospitalisation with Parkinson's disease*

Discrete-time survival with distributed lag models over 10,366,083 Medicare FFS beneficiaries 65+, contiguous US 2000–2016; 10-year exposure histories for PM_{2.5}, NO₂ and summer O₃; non-linear exposure–response. PM_{2.5} **OR 1.634 (1.489–1.792)** at 11.8 vs. 3.0 μg/m³; NO₂ **OR 1.474 (1.379–1.575)** at 31.7 vs. 3.7 ppb; O₃ limited. Associations carried by exposure ≥4 years pre-event. The endpoint is hospitalisation, not incidence — diagnosis-to-hospitalisation lag is unmeasured and plausibly differential by exposure via healthcare access, which is the main threat to the latency claim. PM_{2.5} and NO₂ are strongly correlated in Medicare-scale data and the paper does not resolve which is doing the work.

doi: 10.1097/EE9.0000000000000519

Domingo 2026 Arch Toxicol*Aging in a highly polluted world: challenges and solutions to prevent Alzheimer's disease*

BACKFILL (PubMed entry 2026-06-09; surfaced by the Amass connector). Mechanistic review drawing on human cohorts, animal models and *in vitro* neuronal/glial systems to link pollutant classes — ambient PM, heavy metals, pesticides, PFAS and micro/nanoplastics — to Alzheimer's-relevant neuropathology, framed against the 2024 *Lancet* Commission's inclusion of air pollution among 14 modifiable dementia risk factors and a projected >150 million cases by 2050. Shared pathways discussed are neuroinflammation, oxidative stress, blood–brain-barrier disruption, tau-kinase dysregulation, epigenetic reprogramming and gut–brain dysbiosis, with particular weight on reports of AD-type neuropathology in children and young adults in heavily polluted urban environments. That evidence is the review's most striking claim and also its weakest link: it rests substantially on one research programme's Mexico City autopsy series, where selection into autopsy, APOE distribution and co-exposure to metals and traffic noise are not separable from PM. Read as an argument for exposure reduction as a dementia-prevention lever, it is reasonable; read as evidence of a PM-specific causal pathway, it is not yet that.

doi: 10.1007/s00204-026-04469-x

Parkinsons Dis review 2026 Parkinsons Dis*Air pollution and Parkinson's disease pathology: clinical evidence and molecular mechanisms*

Narrative review linking long-term NO_x, PM and O₃ exposure to Parkinson's incidence and to worse prognosis, with proposed mechanisms spanning gut-microbiome alteration, oxidative stress, neuroinflammation and promotion of α-synuclein aggregation. *No pooled estimate, no systematic search strategy, no risk-of-bias assessment and no quantitative synthesis* — it is a mechanistic orientation piece, not evidence. Tier C accordingly. The gut-microbiome route is the one worth noting, because it implies an ingestion and mucosal pathway for deposited particles that inhaled-dose models do not represent, and because it predicts that the relevant exposure metric is cumulative deposited mass in a specific size range rather than ambient concentration. Carried for the register and for that single mechanistic pointer.

doi: 10.1155/padi/6914112

Sullivan et al. 2026 Dev Psychopathol*Longitudinal impacts of childhood neighborhood air pollution on white matter integrity and cognition*

ABCD Study, *n* = 4,511 children (47.4% female, baseline mean age 9.9), with baseline PM_{2.5} tested against fronto-limbic fractional anisotropy and mean diffusivity across follow-up, controlling for neighbourhood factors. **The direction of association reverses with age:** at 9, higher PM_{2.5} predicted *increased* FA in anterior thalamic radiation, fornix and uncinate; by 14, *reduced* FA in all tracts. In superior longitudinal fasciculus, higher MD at 9 and reduced FA at 12 and 14. **Associations with cognition and substance use were null.** *Limitations:* baseline-only exposure ignores subsequent trajectory; neighbourhood-level assignment carries substantial misclassification; and a sign reversal is equally consistent with accelerated early myelination followed by decline, or with residual confounding that changes with age. Relevance: a design that needs longitudinal, individually resolved exposure and had only a single neighbourhood-level baseline value — the precise gap distributed personal sensing addresses.

doi: 10.1017/S095457942610159X

Xu et al. 2026 Eur J Neurol*PM_{2.5} components and cerebral hypoperfusion, with mediation by inflammatory biomarkers*

967 patients with symptomatic intracranial arterial stenosis at one Suzhou hospital; hypoperfusion quantified from multimodal CT with MISTar; 3-year PM_{2.5} and component exposures from the TAP database; generalized additive models, restricted cubic splines, weighted quantile sum for the mixture, and mediation analysis. **PM_{2.5} OR 4.344 (3.323–5.734); black carbon 4.396 (3.327–5.869)**, then sulfate 4.359, organic matter 4.278, nitrate 3.552, ammonium 3.313 (no intervals given). OM (44.2%) and BC (42.2%) carry 86% of the mixture effect. Mediation by neutrophils 5.37%, leukocytes 4.30%, monocytes 2.73% — i.e. inflammation accounts for roughly a tenth of the total effect, so the dominant pathway is unexplained. Single-centre clinical sample already selected on stenosis; ORs above 4 in that population are not transportable to general-population risk. Component exposures are modelled, not measured.

doi: 10.1111/ene.70716

Respiratory & allergic

11 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (11 OF 11 RECORDS IN THIS SUBTOPIC)

Bogunia-Kubik et al. 2026 Allergy*Influence of environmental pollution on basophil reactivity to birch pollen allergens*

Fifty birch-allergic patients and 25 non-atopic controls followed over the 2023 and 2024 pollen seasons in Kraków, with app-recorded daily nasal, ocular and asthma symptoms against continuously monitored PM_{2.5}, PM₁₀, O₃ and birch pollen. Basophil activation tests used commercial extract, recombinant Bet v 1, and **natural pollen collected from highly and less polluted sites**. Pollen from polluted sites induced significantly stronger CD63 expression ($p < 0.0001$), and asthma symptoms correlated with both pollen and PM ($p < 0.05$, mainly 2023). *Limitations*: the central comparison is site-level, so pollution is confounded with everything else that differs between the two collection sites; $n = 75$ over two seasons with a season-specific correlation is thin; and CD63 is an ex-vivo surrogate. Relevance: if pollution modifies allergen *potency* rather than only co-occurring with it, exposure models that treat pollen and PM as independent additive terms are mis-specified.

doi: 10.1111/all.70474

Feng & Chao 2026 Int J Biometeorol*Lower respiratory infection burden attributable to PM_{2.5} in China and the G20, 1990–2023*

GBD 2023 deaths and DALYs from LRI attributable to ambient and household PM_{2.5}, Joinpoint regression on age-standardised DALY rates with sex subgroups and loess against SDI. Long decline reversed upward in 2021–2023 (APC **China 12.76–14.23%**, G20 **14.94–16.04%** for ambient); household-attributable rates kept falling in China but rose in the G20 from 2021 (APC 8.49–11.20%). Burden concentrated in under-5s and ≥70s. The post-COVID reversal is the interesting claim and it rests on three data-years of GBD modelled estimates, which are themselves smoothed — an inflection this recent should be treated as provisional.

doi: 10.1007/s00484-026-03283-5

Govindaraju et al. 2026 J Occup Environ Med*Protective role of dietary antioxidant intake on long-term effects of extreme PM_{2.5} exposure on respiratory health*

Cross-sectional analysis of 448 Hazelwood Health Study adults, with individual PM_{2.5} from the 2014 Morwell coal-mine fire reconstructed by combining modelled fire-related concentrations with contemporaneous time–location diaries — a better exposure assignment than most disaster epidemiology achieves. Vitamins A and E, magnesium and zinc attenuated the PM_{2.5}–chronic-cough association, and all but magnesium the phlegm association; omega-3 attenuated the PM_{2.5}–COPD association. *Limitations are severe*: diet was measured cross-sectionally years after a 2014 exposure, so reverse causation and healthy-diet confounding are unaddressed; **no effect sizes or intervals are reported in the abstract**; and multiple testing across several nutrients and several outcomes is not accounted for. Relevance: effect-modification analyses of this kind demand exposure precision that only time-resolved personal or time-activity-weighted measurement supplies.

doi: 10.1097/JOM.0000000000003856

The Lancet Respiratory Medicine 2026 Lancet Respir Med*The far-reaching effects of wildfire smoke*

Unsigned editorial. **Abstract not indexed**; carried as metadata only. Logged because wildfire-source attribution has been the recurring through-line of this watch since 27 July and an editorial in this journal signals where the clinical field is placing it. Full text required for any citable content.

doi: 10.1016/S2213-2600(26)00256-0

Makrufardi et al. 2026 World Allergy Organ J*Ambient relative humidity and particulate matter vs. asthma treatment step*

288 newly diagnosed adults with asthma followed at least 6 months under GINA, with treatment step — not symptoms or spirometry — as the outcome, and both daily means and short-term *differences* in RH, temperature and PM_{2.5} as exposures. A 1 µg/m³ increase in the 7-day PM_{2.5} difference was associated with 1.052-fold odds of step 4 treatment and 1.056-fold odds of medium-dose ICS + LABA; a 1 µg/m³ increase in the 30-day difference gave 1.205-fold odds of triple therapy or biologics. Associations were clearer in the cold season. **No confidence intervals are reported in the abstract**, which is why none of these appear in the forest plot. Using variability rather than level as the exposure is the interesting choice — it maps onto what a high-time-resolution sensor measures well and a monthly average does not — but with a single centre and 288 patients this is explicitly hypothesis-generating, as the authors say.

doi: 10.1016/j.waojou.2026.101444

Ni et al. 2026 World Allergy Organ J*Socioeconomic and regional influences on eosinophilic inflammation in chronic rhinosinusitis with nasal polyps*

Multicentre retrospective series, **1,047** CRSwNP patients undergoing endoscopic sinus surgery at three tertiary hospitals in Jiangsu, with paired blood and nasal-mucosa eosinophil counts at 2008, 2012 and 2016, analysed by correlation, ROC, linear regression and linear mixed-effects models. Blood eosinophils are a poor surrogate for tissue eosinophilia ($\rho = 0.28$, **AUC 0.607**, adjusted **R² 0.053**) and vary heterogeneously by hospital and year. City-level GDP correlated with mean blood ($\rho = 0.80$, $p = 0.014$) and tissue ($\rho = 0.72$, $p = 0.037$) eosinophil counts, whereas **annual mean PM_{2.5} was associated with neither**. The null is the reason this record is carried, and it should be read as a measurement failure rather than as evidence of no effect: the exposure is a *city-level annual mean* assigned to surgical patients at three time points, which is an ecological contrast with essentially no within-city variance, over an interval (2008–2016) in which Chinese PM_{2.5} fell steeply and uniformly — so exposure and calendar time are near-collinear by construction, and any true association is absorbed by the year term. With eight city-level observations, the GDP correlations are also fragile. Tier C.

doi: 10.1016/j.waojou.2026.101440

Shi et al. 2026 Medicine (Baltimore)

Air pollution and AECOPD hospitalisation across three Hebei cities

Ecological time series on **14,786 AECOPD hospitalisations** from 49 hospitals in Shijiazhuang, Handan and Chengde, 2018–2020 (1,096 days), with DLNMs adjusted for long-term trend, temperature, humidity, day of week, holidays and pandemic stage, then pooled by random-effects multivariate meta-analysis. Cumulative effects strengthened with lag length and peaked at lag 0–5 d for every pollutant while single-day effects were weak. **PM_{2.5} at the 99th percentile versus median gave RR 1.391 (1.169–1.658) at lag 0–5**; all five non-ozone pollutants showed non-linear cumulative exposure–response, and higher 8-h ozone was associated with *lower* admission risk, which the authors do not explain and which is most plausibly seasonal confounding. The abstract's own conclusions are internally inconsistent on susceptibility — the results state associations were more pronounced in males, the conclusion says females — so the stratified findings should not be cited without the full text. Six correlated pollutants tested across six lag structures with no multiplicity adjustment.

doi: [10.1097/MD.00000000000050080](https://doi.org/10.1097/MD.00000000000050080)

Wang et al. 2026a Front Public Health

COVID-19 social interventions amplified indoor-source pollutant–respiratory associations in Xi'an

Six-year time series of **2,052,777** respiratory outpatient visits over 2192 days at a tertiary hospital in Xi'an (2020–2025), stratified into strict control, transition and post-pandemic phases, with quasi-Poisson GAMs and pollutant-by-phase interaction terms. Full-period associations were strongest for SO₂ (lag 4, ER **9.11%**, 95% CI 3.86–14.64) and NO₂ (lag 5, **2.72%**, 2.05–3.40). Phase-stratified, SO₂ was **18.40% (10.29–27.11)** under strict control versus **2.91% (–4.82 to 11.28)** post-pandemic; PM_{2.5} and O₃ showed no interaction ($p > 0.9$). *The authors state plainly that the interaction tests do not reach significance* ($p = 0.13$, $p = 0.15$), and single-hospital outpatient counts confound exposure effects with healthcare-seeking behaviour that the pandemic also changed. Relevance: the proposed indoor-amplification mechanism is unfalsifiable with ambient monitors alone, which is the argument for indoor and personal measurement in time-series designs.

doi: [10.3389/fpubh.2026.1896806](https://doi.org/10.3389/fpubh.2026.1896806)

Wu et al. 2026 J Thorac Dis

Air pollution and COPD-related health outcomes: bibliometric and visualisation analysis

11,167 Web of Science articles and reviews (1964 to 31 May 2025) through CiteSpace, VOSviewer and bibliometrix: country/institution collaboration, journal co-citation, keyword clustering, burst detection. USA, China and UK lead output; Harvard most influential institution; China and climate change identified as 2021-onward frontiers. Bibliometrics measure publication behaviour, not evidence — useful only for orienting a literature search or a grant narrative.

doi: [10.21037/jtd-2026-0784](https://doi.org/10.21037/jtd-2026-0784)

Yang et al. 2026 J Thorac Dis

COPD and LRI burden attributable to household air pollution and secondhand smoke, 1990–2021

GBD 2021 across 204 countries with BAPC and ARIMA projection. Age-standardised COPD DALYs and mortality fell 37.00% and 37.12%; LRI fell 66.35% and 53.62%. Burden stays concentrated in LMICs, in older women for COPD and in under-15s and ≥60s for LRI; HAP-attributable COPD declined faster than SHS-attributable. Projected female COPD burden from both sources exceeds male by 2030. Standard GBD secondary analysis — the exposure is a modelled solid-fuel-use proxy, not measured PM, and the projections inherit every assumption in the parent estimates.

doi: [10.21037/jtd-2026-0284](https://doi.org/10.21037/jtd-2026-0284)

Zhai & Meng 2026 Int J Environ Health Res

Apparent temperature, air pollutants and pneumonia hospitalisation in rural Western China

Distributed lag non-linear models on NRCMS surveillance data from Jiuquan, 2015–2020, with stratification and explicit pollutant × temperature interaction terms. Apparent temperature showed an inverted-U with a peak **RR 1.50 (1.41–1.60)**. Per-IQR increases gave **CO 1.11 (1.01–1.21)** and **PM₁₀ 1.05 (1.00–1.12)** as the only intervals excluding or touching unity; PM_{2.5} **1.04 (0.97–1.12)**, SO₂, O₃ and NO₂ were null. Synergy was reported between low apparent temperature and high pollution, and among pollutants. Risks were higher in children under 7 and in autumn/winter. A rural setting is a genuine gap-filler — most Chinese DLNM work is urban — but with six correlated pollutants and an interaction screen the multiple-testing burden is not addressed, and the PM findings are the weakest of the set.

doi: [10.1080/09603123.2026.2718327](https://doi.org/10.1080/09603123.2026.2718327)

Reproductive & developmental

5 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (5 OF 5 RECORDS IN THIS SUBTOPIC)

Atalell et al. 2026 Environ Res

Low-level PM_{2.5} and developmental vulnerability at school entry, Western Australia

Whole-of-population linkage of the Australian Early Development Census (2009, 2012, 2015 waves) to midwives' notifications, birth registrations and hospital morbidity data; 54,458 children assigned daily PM_{2.5} at Statistical Area Level 2 across prenatal, early-childhood and cumulative windows; Bayesian multilevel logistic regression against the 2009 10th-percentile domain cut-offs. Mean exposure **7.40–7.54 μg/m³**, below WHO guidance. Early-childhood exposure: **OR 1.04 (1.01–1.07)** for vulnerability in one or more domains and **1.06 (1.03–1.08)** in two or more, per 1 μg/m³, with signals in physical health, social competence and language/cognitive domains. Prenatal and cumulative windows were null. The authors concede the critical limitation themselves: prenatal and early-childhood exposures are highly correlated, so the window attribution is not identified. Area-level assignment at SA2 resolution over a 1 μg/m³ contrast is the second limitation, and the one this watch cares about.

doi: [10.1016/j.envres.2026.125493](https://doi.org/10.1016/j.envres.2026.125493)

Chen et al. 2026 Pediatr Allergy Immunol

Cord blood DNA methylation as a mediator between prenatal air pollution and infant allergic symptoms

Shanghai prospective birth cohort, 101 mother–infant pairs with spatiotemporally modelled residential exposure, caregiver-reported cumulative allergic symptoms to 6 months, and targeted bisulfite sequencing of naïve CD4⁺ T cells in a subset of 30 cord-blood samples. First-trimester PM (**aOR 2.27, 1.09–4.98**) and NO₂ (**2.11, 1.02–4.53**) were associated with allergic symptoms; third-trimester sulfate, nitrate and ammonium components were positively associated; a CpG in the *PRKACB* promoter showed statistical evidence of mediation for the sulfate association, supported by functional assays. The mediation arm rests on 30 samples and the outcome is caregiver-reported at 6 months, which is the weakest available allergic phenotype. The authors label the findings exploratory, correctly. Carried for the component-resolved exposure design, not the effect sizes.

doi: [10.1111/pai.70424](https://doi.org/10.1111/pai.70424)

Hahad et al. 2026 J Environ Health Sci Eng

Air pollution and adverse birth outcomes: a narrative review of epidemiological and mechanistic findings

BACKFILL (PubMed entry 2026-06-29; surfaced by the Amass connector). Narrative review synthesising epidemiological and mechanistic evidence linking prenatal air pollution to low birth weight, preterm birth and stillbirth, with searches across PubMed, Scopus and Web of Science. The two quoted estimates are a **−11.8 g** birth-weight difference and a **+23.1 %** increase in preterm-birth risk per **10 $\mu\text{g}/\text{m}^3$** second-trimester $\text{PM}_{2.5}$; mechanisms discussed are oxidative stress, inflammation, endocrine disruption, vascular dysfunction and epigenetic modification acting on placental morphology, perfusion and metabolic function. Both estimates are quoted **without confidence intervals or provenance to a specific pooled analysis**, which is why neither enters the forest plot — a narrative review is not a meta-analysis, and a point estimate lifted from one without its interval is not an effect size. The review is honest about what remains unresolved: disentangling individual pollutants, establishing causality, and identifying critical windows. Relevant to sensing because trimester-specific windows demand exposure resolution that residential-address annual means cannot deliver.

doi: 10.1007/s40201-026-00990-4

Hung et al. 2026 iScience

Late-pregnancy air pollution, gestational hypertension and neonatal anthropometry

Women recruited in the second trimester and classified gestational-hypertensive or normotensive, with $\text{PM}_{2.5}$, PM_{10} , NO_2 , SO_2 , CO and O_3 over the 30 days before delivery assigned from ambient monitors, and maternal peridelivery hs-CRP, IL-1 β and TNF- α plus cord TNF- α as candidate mediators. GH women had higher hs-CRP and delivered lighter neonates with smaller chest and abdominal circumferences; PM , NO_2 , SO_2 and CO were **inversely associated with birth length**. **Mediation analysis found no significant indirect effect through any measured cytokine**, which is the useful negative — the inflammatory-mediation hypothesis is the default assumption in this literature and it did not hold here. Ambient monitor assignment over a 30-day window is coarse for a length endpoint measured in centimetres, no sample size or effect sizes appear in the abstract, and a single peridelivery cytokine draw is a poor proxy for a gestation-long inflammatory exposure.

doi: 10.1016/j.isci.2026.117029

Wang et al. 2026 Environ Pollut

$\text{PM}_{2.5}$ modifies the temperature–semen quality association

Prospective cohort with repeated semen analysis, linear mixed models with natural cubic splines for temperature, then piecewise regression stratified into higher and lower $\text{PM}_{2.5}$ groups across the low- and high-temperature segments. Temperature showed **inverted-U associations with semen quality over the 0–90, 0–9 and 10–14 d windows** — 0–90 d spans a full spermatogenic cycle, 10–14 d the epididymal transit, so the window structure is biologically coherent. **Synergy between high temperature and $\text{PM}_{2.5}$, antagonism between low temperature and $\text{PM}_{2.5}$** , both acting mainly on progressive and total motility. Stratifying at a $\text{PM}_{2.5}$ median is a crude interaction test that discards within-stratum contrast, and antagonism at low temperature is at least as easily explained by seasonal confounding (heating-season composition, behaviour, indoor time) as by a joint biological mechanism. No effect sizes or intervals are reported in the abstract, so nothing from this record enters the forest plot.

doi: 10.1016/j.envpol.2026.128910

Mechanistic toxicology

11 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (11 OF 11 RECORDS IN THIS SUBTOPIC)

Ayala et al. 2026 Environ Pollut

Contrasting neurotoxic pathways triggered by PM_{10} and organic molecular markers, Catalonia

Fifty-four PM_{10} filter samples collected through 2022 at a rural background site (Bellver de Cerdanya), a suburban site (Manlleu) and a suburban-industrial site (Mollet del Vallès), analysed by GC-MS for 30 organic tracers and tested on SH-SY5Y neuroblastoma cells across six endpoints (viability, ROS, AChE activity, antioxidant and xenobiotic response, p53). MCR-ALS on the joint chemical-biological matrix resolved four components: winter biomass burning (levoglucosan, dehydroabietic acid, PAHs) driving cytotoxicity, oxidative stress and xenobiotic response; a year-round traffic component; a spring–summer SOA component producing **selective AChE inhibition without cytotoxicity**; and a summer POA component. PLS models were significant for five of six endpoints (R^2_{CV} 0.53–0.75, best for ROS, p53 and cell death). The strength is that exposure and effect were measured on the same extracts; the limits are one year, three sites, one immortalised cell line and organic-solvent extracts that do not represent inhaled particle dosimetry.

doi: 10.1016/j.envpol.2026.128961

Chen et al. 2026 Allergy

Gut microbiota signatures reflect $\text{PM}_{2.5}$ -exacerbated asthma in a murine model

Metadata-only: a research letter deposited without an abstract, so nothing below the title is verifiable from the record. Registered here because the gut–lung axis in PM-exacerbated allergic airway disease is a live mechanistic question and because the author group (Academia Sinica Air Quality Research Center with NTU clinical immunology) pairs characterised ambient aerosol with a murine model, which is unusual. No finding, effect size or exposure characterisation is claimed by this issue, and the record supports no figure or statistic in the analytics. Flagged for retrieval once the full text is indexed.

doi: 10.1111/all.70485

Cho et al. 2026 J Microbiol Biotechnol

Lactacisibacillus paracasei ATG-E1 alleviates particulate-matter-induced airway inflammation in association with altered inflammatory signalling, gut microbiota and faecal metabolites

BALB/c mice challenged intranasally with PM_{10}D (PM_{10} plus diesel exhaust particles) on days 0, 3, 6 and 8, then treated orally with ATG-E1 or dexamethasone after inflammation was established — a therapeutic rather than prophylactic design, which is the methodological point in its favour. ATG-E1 reduced BALF inflammatory infiltrate and neutrophilia, lowered collagen deposition and pro-inflammatory mediators, and attenuated **I κ B α and ERK phosphorylation** while leaving JNK and p38 unchanged; caspase-1 and IL-1 α fell, lung RNA-seq showed broad downregulation of cytokine–cytokine-receptor signalling, and cecal 16S plus faecal metabolomics showed microbiome remodelling with increased branched-chain fatty acids and decreased branched-chain amino acids. Two limits: the microbiome and metabolite changes are correlative, with no germ-free, antibiotic or transfer arm to establish that the gut axis mediates the airway effect rather than co-occurring with it; and no effect size, n per group or variance is given in the abstract, so the magnitude is unassessable. All authors are employed by the company that owns the strain.

doi: 10.4014/jmb.2604.04029

Choi et al. 2026 Biofabrication*Degradable porous PLGA/PCL membrane enabling a lung alveoli-on-a-chip for particulate injury*

Porogen-assisted phase-separation membranes with tunable degradation ($\sim 40\%$ surface porosity gain, $\sim 3\ \mu\text{m}$ thickness loss over culture) carrying primary human alveolar epithelium and microvascular endothelium at an air-liquid interface under breathing-mimetic cyclic strain. Apical diesel PM produced dose-dependent cytotoxicity, barrier permeability increase, elevated ROS and DNA damage *in both layers* — trans-barrier propagation, which an insert-only model cannot show. Roflumilast-N-oxide (PDE4 inhibitor) selectively attenuated oxidative and inflammatory readouts without restoring barrier integrity. Deposited dose is delivered apically in suspension and is not equivalent to inhaled dosimetry; no dose-response anchoring to ambient concentrations is attempted. Relevant here as a bench receptor for source-resolved particle samples.

doi: 10.1088/1758-5090/ae9402

Gomes et al. 2026 Cardiovasc Toxicol*Senolytics reverse PM_{2.5}-induced cell and vascular dysfunction*

C57BL/6J mice inhaling concentrated ambient particles versus filtered air showed endothelial activation *in vivo* — increased leukocyte rolling on and adhesion to intact vasculature — alongside senescence of peripheral blood mononuclear cells and endothelial progenitor cells. Dasatinib + quercetin reversed the senescence readouts (β -galactosidase activity, SASP gene expression), restored EPC tube-forming capacity *in vitro*, and limited *in vivo* endothelial activation. Positions senescence upstream of early atherogenesis rather than as a late consequence. Concentrated ambient particles deliver a supra-ambient dose; the senolytic rescue demonstrates sufficiency of the senescence pathway under that dose, not its necessity at ambient concentrations. No exposure-response and no female animals reported.

doi: 10.1007/s12012-026-10171-2

Ichinose et al. 2026 J Hazard Mater*Time-dependent BALT formation and pulmonary inflammation induced by different particulate matter types*

Murine time-course comparing TiO₂ nanoparticles, diesel exhaust particles and Asian sand dust, with and without ovalbumin. All three supported bronchus-associated lymphoid tissue formation, more strongly with allergen. With allergen, an early IL-1 α rise preceded type-2 responses and BALT expansion; **without allergen, BALT still formed via CXCL1 and neutrophil accumulation**, most clearly after repeated TiO₂ — two distinct routes to the same structure. 3D reconstruction showed tertiary lymphoid structures extending from large airways distally, and TiO₂ was localised in alveoli, alveolar macrophages, BALT and hilar lymph nodes. *Limitations*: instilled or high-dose murine exposure is not ambient inhalation; no dose-response to ambient concentrations; and the three particle types differ in dose, size and surface chemistry simultaneously. Relevance: particle *identity*, not mass, determines which inflammatory route is taken — the standing mechanistic case for composition-resolved monitoring.

doi: 10.1016/j.jhazmat.2026.143239

Mahmutovic Persson et al. 2026 Environ Res*Chronic aerosolized PM_{2.5}, plasma proteins and coronary S1P response in rats*

Well-characterised urban PM_{2.5} was re-aerosolised and delivered to rats under two regimes designed to separate an acute high dose from a chronic lower dose at comparable cumulative burden, with coronary vasomotor tone measured *ex vivo* by wire myograph and systemic effects read from plasma sphingosine-1-phosphate and endothelin-1 plus immunohistochemistry. **Only the chronic regime moved anything**: altered S1P vasomotor response in coronary arteries and significantly decreased plasma total protein and ET-1. The design is the point — most PM inhalation toxicology uses acute high-dose exposure because it is tractable, and this says that route can miss the effect entirely. Re-aerosolised collected PM is not fresh ambient aerosol (semivolatiles are lost, morphology changes), the ET-1 *decrease* is not obviously in the direction the vascular-dysfunction hypothesis predicts and is not reconciled, and no n or dose figures appear in the abstract.

doi: 10.1016/j.envres.2026.125391

Seymore et al. 2026 Toxicol Appl Pharmacol*Gestational nanoparticle inhalation alters placental glucose transporters and increases glucose flux*

Pregnant Sprague Dawley rats exposed by whole-body inhalation to TiO₂ nanoparticles at $9.8 \pm 1.0\ \text{mg}/\text{m}^3$, an occupational-level concentration, from gestational day 5 to 19; GLUT1/3/4 assayed in term placenta at GD20 and *ex vivo* placental perfusion used as a functional test. Glut3 mRNA and GLUT1 protein fell, but GLUT4 expression and membrane localisation of both GLUT1 and GLUT4 rose, and perfusion confirmed **increased** glucose flux (AUC 95% CI 77.4–127.5 exposed vs 39.1–73.6 control). The authors state plainly that this contradicts their hypothesis. Read as a compensatory adaptation rather than a benign finding — upregulated placental glucose delivery is not obviously protective in a fetal-growth-restriction model. The dose is roughly three orders of magnitude above ambient PM_{2.5} and the particle is a single engineered material, so this constrains occupational nanomaterial exposure and says little directly about ambient PM.

doi: 10.1016/j.taap.2026.117996

Wei et al. 2026 J Hazard Mater*Seasonal source shifts drive oxidative potential of size-resolved particulate matter in coastal southeastern China*

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar. Listed, not summarised. Noted because size-resolved oxidative potential is one of the few routinely measured quantities that is both a particle property and a plausible biological effect metric, and because a source-driven *seasonal* shift in OP at fixed mass concentration would be direct evidence against mass as a sufficient exposure metric — which is the substantive question behind this watch's instrumentation axis. Whether the paper supports that reading is unknown until an abstract appears.

doi: 10.1016/j.jhazmat.2026.143237

Yuan et al. 2026 Innovation (Camb)*Hallmarks of lung cancer driven by inhalable particulate matter*

Critical review proposing a 12-feature “hallmarks of PM-associated lung cancer” framework spanning cell-intrinsic traits (genetic alteration, epigenetic reprogramming, metabolic remodelling, stemness) and microenvironment remodelling (inflammation, immune suppression, angiogenesis, pre-metastatic niche). Argues for integrating PM exposure into screening and risk stratification. A framework paper: the hallmark taxonomy is an organising device, not evidence, and the underlying studies are heterogeneous in particle source and dose. Useful as a citation scaffold for a mechanism section.

doi: 10.1016/j.xinn.2026.101347

Zhang et al. 2026 J Thorac Dis*Pleural anthracosis severity and the structure and function of subcarinal lymph nodes*

Thoracoscopic grading of pleural anthracosis into four grades by three thoracic surgeons with ImageJ area quantification, against paired subcarinal lymph nodes stained with HE, Masson, EVG and immunohistochemistry for Ki67, CD31, CD68 and D2-40. Black particulate accumulation in node capsule and medulla rose with anthracosis grade, but every quantitative immunostaining index **rose then fell** with advancing severity — a consistent non-monotonic pattern across proliferation, vascularity, macrophage presence and lymphatic integrity. The non-monotonicity is the finding and is also the weakness: with a graded surgical sample and no exposure history, an exhaustion interpretation and a selection interpretation are equally consistent with the data.

doi: 10.21037/jtd-2026-1274

Occupational & indoor**9 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (9 OF 9 RECORDS IN THIS SUBTOPIC)

Alqahtani et al. 2026 PLoS One*Biomarker-based assessment of tobacco smoke exposure in Saudi females via urinary cotinine*

LC-MS/MS quantification of urinary cotinine in 133 Saudi women recruited May–December 2024, validated to ICH-M10 (linearity 2–500 ng/mL, $R^2 \geq 0.9992$, accuracy 90–107%, CV <10%), with a population-specific cut-off of **7 ng/mL**. Cotinine was detected in **48 of 133** samples although **88.7% self-reported as non-smokers**, and 45.9% reported household second-hand smoke. *Limitations:* $n = 133$ is small for establishing a population cut-off; the detected range (0.02–7.693 ng/mL) sits almost entirely *below* the proposed 7 ng/mL threshold, so the cut-off classifies nearly every detected sample as non-smoking and its derivation is not defensible from the reported distribution; and cotinine's short half-life indexes recent exposure only. Relevance: a clean demonstration that self-report and biomarker diverge sharply under social desirability pressure — the same argument for objective personal measurement over questionnaires that motivates wearable PM sensing.

doi: 10.1371/journal.pone.0355166

Herkert et al. 2026 Environ Sci Technol*Contribution of indoor air and dust to children's PFAS exposure in the Faroe Islands*

Measurement of nonvolatile precursors, volatile neutral precursors and legacy ionic PFAS in air and dust across **40 Faroese homes**, with estimated daily intake computed for 15-year-olds. EDI ranges 0.001 / 0.012 / 0.129 ng/kg bw across low, intermediate and high scenarios, equal to roughly **0.2%, 2% and 20% of the EFSA tolerable weekly intake**, against ~60% from pilot-whale consumption. Indirect exposure to perfluoroalkyl carboxylic acids via air dominates, followed by PFOS in dust. *Limitations:* scenario-based EDIs are not measured intakes, and the 40-home sample cannot support the tail of the distribution where the 20% figure sits. Relevance: the indoor compartment supplies a fifth of a regulatory reference dose in the high scenario for a population with no local source — a direct case for indoor sampling as a routine part of exposure assessment rather than a specialist add-on.

doi: 10.1021/acs.est.6c02003

Iavazzo et al. 2026 Environ Geochem Health*Particle characterization in an urban synthetic turf: turf and vehicular wear*

XRF, SEM-EDS and dynamic light scattering on dust, turf fibre and crumb-rubber infill from an urban synthetic pitch, with stub-based sampling worn on players. Irregular and elongated fine, submicron and nano-sized particles were observed; Fe dominated by XRF, with Fe, Si, Al and Br-bearing particles by SEM-EDS and **Ni, Ag and Rh** recovered from the player-worn stubs — Rh being a clear three-way-catalyst tracer, so the pitch is receiving vehicular as well as turf-wear particles. Ambient PM_{2.5} averaged **10.93 µg/m³** (24-h) and O₃ reached 41.12 ppb. *Limitations:* one field, no personal inhaled dose, and the deterministic USEPA assessment — which found no significant non-carcinogenic metal risk — is mass-based and cannot speak to nanoparticle toxicity. Relevance: the contrast between an unremarkable regulatory 24-h mean and a demonstrably loaded microenvironment is the clearest argument in this issue for personal and microenvironmental sampling.

doi: 10.1007/s10653-026-03407-1

O'Donnell et al. 2026 medRxiv (preprint)*Nicotine replacement therapy to reduce children's second-hand smoke exposure at home*

Pilot randomised controlled trial in Scotland testing whether free NRT plus telephone-delivered support is a feasible route to a smoke-free home, with objective indoor PM_{2.5} as the exposure measure rather than self-reported smoking — which is the design decision that makes it relevant here. Feasibility, recruitment and retention are the outcomes; the trial is not powered for exposure reduction. Notable as one of very few interventions targeting the indoor source directly rather than filtering downstream of it. Preprint, not peer reviewed; no numeric claim from it is used in this issue's analytics.

doi: 10.64898/2026.08.13.26360348

Ranjbar et al. 2026 Preprints.org (preprint)*1-Nitropyrene as a diesel-specific indoor exposure proxy in California homes*

1-Nitropyrene and the isomeric 2-nitropyrene and 2-nitrofluoranthene measured in indoor air and house dust from **40 homes** in the eastern San Francisco Bay Area across two sampling events, alongside continuous indoor black carbon. The motivation is specificity: BC, NO_x and PM_{2.5} are all used as diesel surrogates and none is diesel-specific. Detection was **74% in air and 97% in dust**, medians **0.42 pg/m³** (IQR 0.32–0.54) and **340 pg/g** (IQR 190–620); air–dust correlation $r = 0.40$ ($p = 0.02$) and BC–1-NP $r = 0.38$ (air) and 0.47 (dust). Diagnostic ratios (2-NFl:1-NP < 5) indicate primary emission rather than secondary formation. Dust 1-NP associated with more spatial diesel predictors than air 1-NP did, consistent with dust integrating longer-term deposition — which is the actual argument for using it. Preprint, not peer reviewed. Forty homes, two events, no personal sampling, and a picogram-per-cubic-metre analyte whose blank and recovery performance is not summarised in the abstract.

doi: 10.20944/preprints202608.0873.v1

Salcines et al. 2026 J Environ Health Sci Eng

Monitoring and risk assessment of airborne graphene oxide released during 3D printing with polylactic acid filaments

Tiered exposure assessment to EN 17058:2018 and OECD guidance around fused-deposition printing of graphene-oxide-loaded PLA, combining **near-field and far-field particle counters** with SEM and Raman speciation plus VOC measurement against Spanish and Finnish 2022 occupational limits, with ARIMA used to test whether the observed excursions are distinguishable from background drift. Nanoparticles and agglomerates were present at low average number concentration and **GO was detected in the respirable fraction**, below proposed limits but demonstrably released — the finding is qualitative (release occurred) rather than quantitative (how much). Weaknesses are structural: a pilot with no stated replicate count, number-concentration monitors that cannot speciate, so the GO attribution rests entirely on offline SEM/Raman of collected material, and occupational limits for GO that do not formally exist. The methodological point transfers directly — paired near-field/far-field counters with an offline speciation leg is the minimum viable design for attributing a number-concentration excursion to a source, and it is the design a distributed indoor network would need.

doi: 10.1007/s40201-026-00998-w

Shadmani et al. 2026 J Environ Health Sci Eng

A comprehensive health profile and exposure-risk analysis in an iron ore mining workforce

Cross-sectional study of **1,193** Iranian iron-ore mine employees, stratified into high, moderate and low exposure *by job role*, with physical examination, clinical chemistry and haematology compared by ANOVA and χ^2 and modelled by linear and logistic regression. The workforce is young (mean **34.9 ± 7.6 y**) with low chronic-disease prevalence; the burden is localised — hearing **23.8 %**, vision **17.8 %**, dermatological **31.3 %** — while most haematological and metabolic markers did not differ by exposure group. Platelet count, LDL and blood pressure were *lower* in the high-exposure group, which the authors do not resolve and which has an obvious competing explanation the paper does not exclude: the healthy-worker survivor effect plus role-based confounding by age and seniority. The central methodological weakness for this watch is that **exposure is a job-role proxy with no measured dust concentration at all** — no gravimetric respirable sampling, no personal monitoring, no size fractionation. A three-level ordinal surrogate cannot support an exposure–response claim, and the paper’s own conclusion (localised sensory and dermatological effects, not systemic metabolic ones) is a statement about a young cohort with short cumulative exposure rather than about the hazard.

doi: 10.1007/s40201-026-01000-3

Tan et al. 2026 Nutrients

PM_{2.5} components, nutritional indices and leisure-time physical activity vs lower eGFR in subway workers

8,477 Wuhan subway workers; PM_{2.5} and components from the Tracking Air Pollution in China dataset; creatinine-based eGFR; GNRI and PNI nutritional indices; IPAQ-derived activity. Mixture effects by WQS, quantile g-computation and BKMR. **WQS OR 1.183 (1.106–1.266) and QGC OR 1.183 (1.083–1.291)** for lower eGFR; BKMR overall effect on eGFR of -0.045 ($-0.068, -0.022$) for a 5-percentile mixture increase against medians. Subgroups by nutrition and activity differed but **no significant interaction** was found — the authors say so, and the correct reading is hypothesis generation, not effect modification. Cross-sectional, so reverse causation is unaddressed; and residential exposure is a poor proxy for the occupational exposure of people who work underground, which is the population’s defining feature.

doi: 10.3390/nu18152580

Zhao et al. 2026 J Int Med Res

Occupational dust exposure and thyroid nodule screening: a predictive modelling approach for early detection in coal mining communities

Cross-sectional analysis of **1,708** coal miners examined at an occupational-disease hospital in Anhui in April 2021, combining general, laboratory and imaging data to predict nodular thyroid disease, using a proposed “Nonlinear Inverse Nearest Manifold Projection” model benchmarked against established risk models. The record is carried at tier C and should not be cited for a result: the abstract reports that the model “achieved high scores in both F1 and area under the precision–recall curve” and “significantly outperformed” comparators **without giving a single numeric value**, confidence interval, or comparator name. Beyond the missing numbers, dust exposure is never measured — occupation is the exposure proxy, with no mass concentration, no respirable fraction, no duration and no job-exposure matrix — so the paper cannot support any dose–response claim about particulate exposure and thyroid nodules. Included because occupational particulate exposure with thyroid endpoints is a thin literature and the register should show what is in it.

doi: 10.1177/03000605261473853

Burden, policy & mitigation

15 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (15 OF 15 RECORDS IN THIS SUBTOPIC)

He et al. 2026 Neuroepidemiology

Premature ischaemic stroke burden in adults aged 50–69 in China, India and the United States

GBD 2023 DALYs, YLLs, YLDs and population attributable fractions for the 50–69 stratum in three countries at different stages of epidemiological transition, with ARIMA projection to 2053. Trajectories diverge: China declining, India flat at a high level, the US on a high plateau; the US has shifted to a YLD-dominant (disability) composition while India remains YLL-dominant (acute mortality); China shows a near 2:1 male:female ratio driven by male smoking. **India is characterised as facing dual pressure from persistent PM_{2.5} and rising metabolic risk**. The authors are explicit that widening intervals make the 2053 horizon scenario-based rather than forecast; a 34-year ARIMA extrapolation from modelled burden estimates is presentation, not evidence. Carried for the PAF decomposition only.

doi: 10.1159/med/adwag005

Lee et al. 2026 J Korean Med Sci

Toward a unified framework for climate and health research: gaps in exposure metrics and health outcomes

Mapping review of **611** original human studies (PubMed, Cochrane, Embase; January 2000 – June 2024) on climate-related exposures and health. The distribution is sharply skewed: **526 studies on temperature and temperature-related hazards, 88 on precipitation, and only 39 on air pollution**, with extreme weather barely represented. Outcomes concentrate in all-cause mortality (249), cardiovascular disease (159) and communicable disease (148), while neurodegenerative disease (8) and endocrine disorders (4) are almost absent. Exposure-metric definitions are described as substantially heterogeneous. *Limitations*: counts are of studies, not of evidence quality, and English-only inclusion biases the geography. Relevance: the two endpoints this issue adds — thyroid function and fronto-limbic white matter — sit in the two thinnest outcome categories the review identifies, and the metric heterogeneity it documents is the same problem that makes cross-study pooling of PM exposure estimates unreliable.

doi: 10.3346/jkms.2026.41.e294

Atmos Environ 2026d METADATA ONLY Atmos Environ*Drivers of air pollutant and greenhouse gas emissions since 2010 in the Greater Bay Area, China*

METADATA ONLY — no abstract retrievable at build time. Carried for the register because decomposition of co-emitted air pollutants and greenhouse gases over a 15-year window in a single megacity cluster is the analysis that determines whether observed PM improvement came from control technology, fuel switching, activity shift or economic structure — and those four have different implications for what happens next. Method, pollutant coverage and decomposition scheme are all unstated. Lowest-weight record in the issue.

doi: 10.1016/j.atmosenv.2026.122301

Liu D. et al. 2026 Medicine (Baltimore)*COPD burden 1990–2050 with risk-factor attribution across age and SDI*

GBD 2021 across 204 countries, analysed by joinpoint regression, age–period–cohort models and decomposition into population growth, ageing and epidemiological change, with Bayesian APC projection to 2050. Age-standardised rates fell (AAPC -0.05% incidence, -1.49% mortality, -1.48% DALYs) while absolute burden rose to **213.4 million cases and 1.69 million deaths in 2021. Ambient $PM_{2.5}$ carries 15–25% of DALYs and household air pollution 0–55%, against smoking at 38–55%, with an inverted-V SDI relationship peaking at SDI 0.4–0.6. The household-air-pollution range spanning zero to fifty-five percent is the whole story and is presented as a single interval; it is a statement about between-region heterogeneity, not uncertainty, and the paper does not separate the two. GBD risk-factor PAFs inherit their exposure–response functions from the integrated exposure–response curves, so this is a re-analysis of GBD outputs rather than independent evidence. Tier C accordingly.**

doi: 10.1097/MD.00000000000050162

Liu M. et al. 2026 Environ Monit Assess*Divergent near-surface ozone and NO_2 trends and ozone-attributable burden, Huaihe River Basin*

Twelve years (2013–2024) of high-resolution satellite-derived surface $MDA8\ O_3$ and NO_2 across the Huaihe Basin, with pixel-wise OLS trends, the Hurst exponent as a persistence diagnostic, and BenMAP-CE for attributable premature mortality and economic loss. **O_3 rose at $2.792\ \mu g\ m^{-3}\ yr^{-1}$ (mean $104.15\ \mu g\ m^{-3}$) while NO_2 fell at $-1.385\ \mu g\ m^{-3}\ yr^{-1}$ (mean 29.86)** — the textbook VOC-limited response to NO_x -only control, here quantified at basin scale with an outward expansion of O_3 -affected area and contraction of NO_2 hotspots. Cardiovascular mortality was the largest cause-specific component, concentrated in Jiangsu and Shandong. **No PM metric is estimated**, which for a basin whose $PM_{2.5}$ and O_3 share precursors is a real gap; and the Hurst exponent applied to a 12-point annual series is a weak persistence estimator being asked to carry a forward-looking claim.

doi: 10.1007/s10661-026-15802-3

Liu et al. 2026b Front Oncol*Lung cancer incidence and mortality trends with age-period-cohort analysis, Zibo 2014–2024*

Registry data for a heavily industrial Shandong city. Age-standardised incidence rose (AAPC **$+4.34\%$** , 95% CI 2.14–6.59) while mortality fell (AAPC **-2.13%** , -3.53 to -0.72); male rates exceeded female throughout. Period effect increasing, cohort effect decreasing across successive birth cohorts. Rising incidence with falling mortality in a screening-active decade is the classic overdiagnosis signature and the paper does not test it; air pollution appears as narrative context with no exposure measurement at all.

doi: 10.3389/fonc.2026.1873592

Mou et al. 2026 Environ Pollut*Ship emissions and regional air quality in the post-DECA era, Yangtze River Basin*

A bottom-up 2023 emission inventory for **18,478 vessels** built from **≈ 230 million AIS trajectories** on the world's busiest inland shipping corridor, coupled to WRF-Chem for concentration attribution. SO_2 and $PM_{2.5}$ ship emissions are substantially controlled post-DECA, but **NO_x reached $156.41\ kt\ yr^{-1}$, 82.0% of major-pollutant mass**, with hotspots at Yichang–Jingzhou, Wuhan and Nanjing–Shanghai. Ship contribution to NO_2 was **0.16 – $3.41\ \mu g\ m^{-3}$** annual mean across 11 mainstem cities; $PM_{2.5}$ increments were small but nitrate-dominated; O_3 fell in January and October under near-source NO titration and rose in July. The policy read is clean — fuel-sulfur control has done its work and the residual problem is NO_x — but the inventory rests on activity-based emission factors with no plume measurement to validate them, and the $PM_{2.5}$ increment excludes secondary nitrate formed downwind of the model domain.

doi: 10.1016/j.envpol.2026.128946

Okeke et al. 2026 Environ Monit Assess*Agricultural methane emissions and non-communicable disease mortality in West Africa*

Ecological panel of 11 West African countries, 2015–2020, relating agricultural methane (Mt CO_2e , World Bank WDI) to the probability of dying from NCDs between ages 30 and 70, using pooled OLS, country fixed effects and mixed models, with The Gambia (-28.3% emissions) as a descriptive contrast. *This is a weak design and should be read as such:* 11 countries $\times$ 6 years is 66 country-years, the exposure is a national accounting aggregate with no measured concentration and no plausible direct mechanism to NCD mortality, and the outcome is a modelled WHO indicator rather than observed deaths. Ecological fallacy and confounding by development trajectory are unaddressed. Tier C, carried only because it is one of the few Sub-Saharan records in this watch and marks where the climate–health literature is expanding without measurement infrastructure — which is itself the finding.

doi: 10.1007/s10661-026-15800-5

Sathasivam & Veluchamy 2026 Environ Sci Pollut Res*Visual characteristics of PM from *Halymenia dilatata* microalgae biodiesel compared to low-sulfur diesel*

Single-cylinder Kirloskar CI engine, four microalgae biodiesel blends (B10–B25) against S10 low-sulfur diesel, PM collected on a textile element in a deceleration chamber, quantified gravimetrically and characterised by SEM for morphology, deposition and agglomeration. **PM mass reduction 10.44–69.79%** across blends. The result does not vary monotonically with blend ratio — SEM showed heavier deposition for S10, B15 and B25 but lighter for B10 and B20 — which the paper reports but does not explain, and a non-monotonic dose–response across four points with no stated replication is more likely measurement noise than combustion chemistry. Three further limits matter: a textile collection element in a deceleration chamber is not a dilution tunnel and has no characterised collection efficiency as a function of size, so the gravimetric numbers are not comparable with any regulated emissions measurement; no size distribution or number concentration was measured, so a mass reduction is fully compatible with an increase in ultrafine number; and no engine load or speed conditions are given in the abstract. Included as a tier-C source-characterisation record.

doi: 10.1007/s11356-026-38134-8

Suryanarayanareddy et al. 2026 Environ Sci Pollut Res Int*Low-cost bio-based particle-capturing device for vehicular emissions: pilot study*

Cylindrical coconut-husk and sugarcane-pulp enclosure tapering to a conical outlet and a monitoring chamber, tested indoors under simulated junction conditions at three turbulence intensities. Single-pass removal 15–20% ($PM_{2.5}$) and 15–45% (PM_{10}) for halted vehicles; 20–65% and 15–65% for moving vehicles. Efficiency ordering flips with turbulence regime, which the paper reports honestly. Indoor simulation of an outdoor near-road environment, no replication count given, and no material-degradation testing on a biodegradable enclosure meant to sit at a roadside.

doi: 10.1007/s11356-026-38118-8

Tian et al. 2026 Nanomaterials (Basel)

Bioinspired honeycomb-structured nanofibrous membranes for high-efficiency PM capture

Template-assisted electrospinning onto a mesh receiver establishes a directional electric field, producing concentrated ordered stacking plus sparse random orientation of fine-diameter fibres. **>98.51% filtration efficiency at 34 Pa pressure drop**, 85% transparency, 130.8 mm/s air permeability. Efficiency is quoted for a single challenge condition; no loading curve, no electrostatic-decay ageing, no humidity sensitivity — the three things that determine whether an electret-assisted filter holds its number in service. Bench result, not a service life.

doi: 10.3390/nano16150941

Urso et al. 2026 Lancet Reg Health Eur

Burden of dementia in Europe, 1990–2023: a systematic analysis from the Global Burden of Disease Study 2023

GBD 2023 secondary analysis for the EU-27 and the WHO European Region, with non-fatal outcomes from DisMod-MR 2.1 and dementia-attributable mortality from an excess-mortality framework. **7.76 M (95% UI 6.69–8.72)** prevalent cases in the EU-27 and **12.28 M (10.44–13.90)** across the WHO European Region, both roughly **+90% since 1990** despite modest declines in age-standardised prevalence; dementia moved from the eighth to the **third** leading cause of death in the EU-27. **41% (24.4–57.4)** of EU-27 dementia DALYs are attributed to six modifiable risk factors, with **ambient particulate matter pollution named first**, ahead of high fasting plasma glucose and high BMI, and concentrated in socioeconomically disadvantaged populations. The limitation is the one every GBD risk-factor attribution carries and this paper does not resolve: the PM_{2.5}–dementia relative risk is imported from a comparative-risk-assessment function fitted to a thin and heterogeneous cohort literature, and the **24.4–57.4** interval is a modelling uncertainty, not a sampling one — it does not propagate uncertainty in the exposure surface, which for a 1990–2023 European back-cast is substantial. Treated here as a burden estimate, not as neurological evidence. The interval is deliberately absent from the forest plot: an attributable fraction has no position on a ratio axis.

doi: 10.1016/j.lanepe.2026.101796

Vartanyan et al. 2026 Cureus

Structural inequities and chronic disease in Fresno County: a narrative review and multi-sectoral framework

Comparative descriptive analysis of public-health data for **2018–2021** showing Fresno County, California, exceeding state prevalence and mortality for obesity, hypertension, diabetes, asthma, coronary heart disease and stroke, attributed to poverty, low educational attainment, provider shortages, food insecurity and severe air and pesticide pollution, with disproportionate impact on racial and ethnic minority populations. Tier C, and the reason is structural rather than a criticism of execution: this is an ecological comparison of county aggregates in which air pollution is one of several correlated exposures, none measured at individual level, so no exposure-specific inference is available and none is claimed. Carried because the San Joaquin Valley is among the most PM_{2.5}-burdened airsheds in the United States and because the paper's underlying argument — that monitoring density and exposure burden are inversely related in marginalised communities — is the same one Siu et al. quantified for eastern Canada in the 5 August issue, here asserted rather than measured.

doi: 10.7759/cureus.112418

Wu & Cui 2026 J Glob Health

Aircraft CO, NO₂ and SO₂ emissions and public health at Chinese airports

Landing-and-take-off cycle data from the Civil Aviation Administration of China, concentrations at each airport from a Gaussian diffusion model, premature deaths from the Global Exposure Mortality Model applied to populations near airports. Attributable mortality was higher in men than women and concentrated in older age groups; CO accounted for the most respiratory deaths; the peak burden year was 2019. **No particulate metric is estimated at all**, which for an aviation source — where ultrafine particle number is the exposure of concern and is not proxied by CO — is a substantive omission rather than a scope choice. A Gaussian plume model at airport scale with no measurement validation is the second limitation. Carried at tier C for the burden framing and as a marker of where aviation-adjacent monitoring is not being done.

doi: 10.7189/jogh.16.04210

Xie et al. 2026 J Thorac Dis

Thirty-year change and projection of Asian tracheal, bronchus and lung cancer burden

GBD 2021 across Asian subregions with Joinpoint, age-period-cohort, decomposition, frontier, inequality and BAPC prediction. Incidence roughly doubled and deaths rose ~2.7-fold 1990–2021; ASIR +22%, ASPR +49%, ASMR +8%. Population growth was the largest contributor, then ageing. **Particulate matter pollution ranks second behind smoking among risk factors at ~10% and rising**. Six analytic frameworks on one modelled dataset multiply presentation, not information.

doi: 10.21037/jtd-2026-0740

Other clinical endpoints

11 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (11 OF 11 RECORDS IN THIS SUBTOPIC)

Chin et al. 2026 Res Sq (preprint)

Impact of air quality and PM_{2.5} on hidradenitis suppurativa activity in the Sacramento region

Retrospective analysis of **10,431** daily hidradenitis suppurativa encounters at UC Davis Health, **April 2015–March 2022**, regressing monthly encounter counts on the monthly count of days exceeding **12 µg/m³** PM_{2.5} from the EPA Air Quality System, with lags to three months. Positive associations at **lag 0** ($\beta = 0.5658$, $P = 0.022$) and **lag 2** ($\beta = 0.6230$, $P = 0.015$); lag 1 and lag 3 null. The lag structure is the problem. A significant effect at 0 and 2 months with nothing at 1 month has no biological reading, and four lags tested without adjustment on a monthly series of ~84 points is a multiple-comparisons pattern. Three further limits: healthcare encounters measure care-seeking as much as disease activity and are strongly seasonal; a regional monitor assigned to a referral population is a substantial exposure misclassification; and no autocorrelation correction or seasonal adjustment is described, which for monthly counts over seven years will inflate significance on its own. Preprint, single centre, not peer reviewed.

doi: 10.21203/rs.3.rs-10570042/v1

Chow et al. 2026 Nat Commun

Blood metabolomic signatures linking air pollution to lung cancer

Prospective nested case-control across CPS-II and CPS-3, 1,357 participants, >1,100 metabolites in pre-diagnostic plasma, residential concentrations of six pollutants at blood draw. Eight metabolites associated with both exposure and subsequent lung cancer; four (γ -glutamylglutamine, phenylacetylglutamate, N-(2-furoyl)glycine, 4-vinylguaiacol glucuronide) significantly mediated PM and ozone associations at adjusted $q < 0.2$. Single-timepoint metabolomics cannot separate exposure response from early disease, and $q < 0.2$ is a permissive threshold over 1,100 features — treat the four as candidates. Smoking adjustment in a lung cancer metabolome study is the obvious residual-confounding exposure.

doi: 10.1038/s41467-026-75116-3

Curr Hematol Malig Rep 2026 Curr Hematol Malig Rep*Linking air pollution and hematologic malignancies: from epidemiology to molecular mechanisms*

Review of outdoor air pollution and haematologic malignancy across acute and chronic leukaemias, lymphomas, MDS, CHIP and myeloma, implicating PM, benzene, NO₂, SO₂, arsenic and ethylene oxide through oxidative stress, DNA damage, epigenetic dysregulation, chronic inflammation and haematopoietic stem-cell dysfunction. Evidence is graded strongest for AML with benzene; ALL associations with traffic-related pollution, NO₂ and PM are described as less consistent; evidence for chronic leukaemias and myeloma is limited. *Not systematic*: no search protocol, no pooled estimates, no bias assessment, and the authors themselves flag exposure misclassification and residual confounding. The CHIP link is the forward-looking one — a clonal intermediate would give this literature a measurable early endpoint. Relevance: benzene and PM co-vary in traffic sources, and separating them requires speciated measurement that mass monitors cannot provide.

doi: 10.1007/s11899-026-00785-2

Jain et al. 2026 J Curr Glaucoma Pract*Ambient air pollution and glaucoma: systematic review and meta-analysis*

PRISMA review, PubMed and Embase to 30 December 2025; 18 observational studies, 8 pooled by generic inverse variance. **Pooled RR 1.12 (1.06–1.18)** for long-term PM_{2.5}, $I^2 = 78.7\%$. NO₂, NO_x, PM₁₀ and CO directionally positive but synthesised narratively. RNFL and GCIPL thinning and modest IOP elevation also reported. Heterogeneity at 79% with eight studies makes the pooled point estimate a weak summary; the authors say causality cannot be established and they are right. Two databases only is a real recall limitation for this literature.

doi: 10.5005/jp-journals-10078-1522

Li et al. 2026 Innovation (Camb)*Global and national CKD burden attributable to fine particles: updated exposure–response meta-analysis*

Three-stage mixed-effects meta-analysis over 38 CKD and 16 eGFR studies, fitting a non-linear ERF and propagating it to 1990–2020 burden above a 10 µg/m³ counterfactual. **RR 1.32 (1.21–1.43) per 10 µg/m³**, predominantly log-linear; **eGFR –2.115 (0.452–3.777) mL/min/1.73 m²** with saturation at high exposure. 2020 burden: 485,000 deaths, 5.77 million incident and 243 million prevalent cases. The counterfactual floor is a choice, not a threshold, and the burden figures scale with it; the primary studies are also concentrated in high-exposure Asian settings, so the low-concentration arm of the ERF is thin.

doi: 10.1016/j.xinn.2026.101429

Liu et al. 2026 JACC Adv*Reply: statistical fragility and methodological constraints in metabolomic mediation*

Correspondence from the Emory group (Liu, Chang, Jones, Quyyumi, Liang) responding to a critique of metabolomic mediation between air pollution and cardiovascular risk. **Abstract not indexed** — carried as metadata only because it sits directly against Chow et al. (same senior author, same method class) in today's harvest and the methodological dispute is the reason to read both. Full text required.

doi: 10.1016/j.jacadv.2026.103149

Mandal et al. 2026 Kidney Int Rep*Long-term ambient PM_{2.5} and kidney function in urban Indian adults*

Population-representative cohort of **12,271 adults** aged 20 and over in Chennai and Delhi, with eGFR (CKD-EPI 2009) at three points between 2010 and 2016 and annual mean PM_{2.5} assigned at residential geocode from a 1 km national daily spatiotemporal model. City-specific linear mixed models with inverse-probability weighting for loss to follow-up, plus effect modification by sex, waist-hip ratio, hypertension and diabetes. Exposure contrast between cities is extreme — Chennai 33/37/30 versus Delhi 118/123/130 µg/m³ — and both cities show decline: **–0.32 (95% CI –0.49 to –0.15)** in Chennai and **–0.42 (–0.57 to –0.27) mL/min/1.73 m² per 5 µg/m³** in Delhi. That the two cities agree despite a fourfold concentration difference is reassuring for the shape of the relationship and awkward for its magnitude. CKD-EPI 2009 is poorly calibrated in South Asian populations, three measurements over six years is thin for an eGFR slope, and the by-city stratification means the between-city contrast — the largest exposure contrast in the data — is never used.

doi: 10.1016/j.ekir.2026.106693

Negahi et al. 2026 Int J Biometeorol*Climate-sensitive exposures and cancer outcomes: systematic review and meta-analysis*

Five databases searched to May 2026 for observational studies of wildfire, ambient temperature, hurricane or multi-hazard disaster, precipitation anomaly, or ambient PM_{2.5} against cancer incidence, mortality, survival or recurrence; 25 estimates from 14 studies pooled within ten exposure-outcome groups, with robust variance estimation as the primary method for groups containing dependent estimates. **Uncorrected pooled 1.07 (1.02–1.12); RVE 1.02 (0.93–1.11)** — the conventional estimate was inflated by correlated effects, with $I^2 = 97\%$. Wildfire and cancer mortality **1.11 (0.89–1.38)**; ambient PM_{2.5} associated with poorer survival; an apparently protective temperature-survival association vanished under RVE and is best read as residual confounding. GRADE certainty was low in two groups and very low in the other eight. Fourteen studies across five exposure types and four outcome classes is too thin a base for group-specific inference, and the paper says so.

doi: 10.1007/s00484-026-03293-3

Schlaepfer et al. 2026 Transl Androl Urol*Environmental hazards and the risk of idiopathic urethral stricture*

Utah Population Database, 2,048 cases born 1996–2021 matched 1:1 on birth year and location to 1,939 controls; historical air pollution and toxics exposure assessed prenatally, in the first year of life, and in the five years before diagnosis. **Only the first-year window was associated: RR 1.21 (1.07–1.36)**. Univariate signals for ethylbenzene, tetrachloroethylene and molybdenum trioxide. Three exposure windows tested with one positive is a multiplicity problem the paper does not address, and the univariate toxicant hits are unadjusted. Included as a hypothesis, not a result.

doi: 10.21037/tau-2026-0338

Traisathit et al. 2026 PLoS One*Air pollution and mortality in acute lymphoblastic leukaemia across age groups, northern Thailand*

604 ALL patients diagnosed 1999–2020 in upper northern Thailand — a region with severe seasonal biomass-burning haze — stratified into paediatric (<15, n=344), adolescent and young adult (15–39, n=152) and adult (≥40, n=108). Five-year survival 64.5%, 19.6% and 6.7% respectively; adjusted HR for death 3.76 (AYA) and 8.10 (adult) versus paediatric; diagnosis before 2010 aHR 1.38. **PM_{2.5} ≥50 µg/m³ was associated with increased mortality in paediatric patients**. The age-group HRs are prognosis, not pollution. The pollution claim rests on comparing mean concentrations between those who died and those who did not, with a threshold analysis in one stratum; no exposure window, no treatment-era adjustment beyond the pre/post-2010 term, and ambient station concentrations as individual exposure. Signal, not estimate.

doi: 10.1371/journal.pone.0355856

Wang et al. 2026b J Hazard Mater*Ambient PM_{2.5} and thyroid health: 24,333 participants in Zhejiang, China*

Community-based cross-section of **24,333** participants (2022–2025) spanning minors and adults, with PM_{2.5} assigned from geocoded residential addresses and mixed models estimating associations with FT4, TSH, subclinical hypothyroidism and autoimmune thyroiditis. Per 10 µg/m³: FT4 **–3.57% (95% CI –4.39, –2.75)**, TSH **+12.15% (7.92, 16.56)**, subclinical hypothyroidism OR **1.27 (1.06, 1.51)**; no clear association with autoimmune thyroiditis. The FT4 effect was larger in under-18s and in BMI <18.5; the TSH and hypothyroidism effects larger in those ≥50 and BMI ≥24. *Limitations*: cross-sectional, so exposure precedence is assumed not observed; residential-address assignment ignores time-activity; and iodine status is invoked as a rationale but its adjustment is not described in the abstract. The internally coherent FT4-down/TSH-up pattern argues against pure chance. Relevance: the paediatric and low-BMI susceptibility windows require exposure resolved at the individual rather than the address.

doi: [10.1016/j.jhazmat.2026.143231](https://doi.org/10.1016/j.jhazmat.2026.143231)

Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Delaney et al.	Cumulative annual air pollution exposure and first hospitalization with Parkinson's disease	<i>Environ Epidemiol</i>	Registry cohort + geospatial	PM _{2.5} , NO ₂ , summer O ₃	United States
Domingo	Aging in a highly polluted world: challenges and solutions to prevent Alzheimer's disease	<i>Arch Toxicol</i>	Mechanistic review	Ambient PM + metals, PFAS, micro/nanoplastics	Multi-region
Parkinsons Dis review	Air pollution and Parkinson's disease pathology: clinical evidence and molecular mechanisms	<i>Parkinsons Dis</i>	Mechanistic review	PM, NO _x , O ₃	Global
Sullivan et al.	Longitudinal impacts of childhood neighborhood air pollution exposure on white matter integrity and cognition	<i>Dev Psychopathol</i>	Prospective cohort	PM _{2.5}	United States
Xu et al.	Association between PM _{2.5} components and cerebral hypoperfusion: a mediation role of inflammatory biomarkers	<i>Eur J Neurol</i>	Retrospective clinical series	PM _{2.5} , black carbon, sulfate, nitrate, ammonium, organic matter	China
Cardiovascular & metabolic					
Chen et al.	Compound cold wave-PM _{2.5} exposure and cardiovascular emergency ambulance calls	<i>Research Square (preprint)</i>	Ecological panel	PM _{2.5}	China
Gaggini & Vassalle	Gene-air pollution interaction in cardiovascular disease	<i>Int J Mol Sci</i>	Narrative review	PM _{2.5} , PM ₁₀ , traffic-related pollutants	Global
Kakhaia et al.	Causal effects of time-varying air pollution mixtures on cardiovascular disease incidence	<i>Environ Int</i>	Prospective cohort (mixtures)	PM _{2.5} , NO ₂	United Kingdom
Moro et al.	Long-term ambient PM _{2.5} exposure, hypertension and metabolic syndrome among Ghanaian adults	<i>Environ Res</i>	Cross-sectional (cohort baseline)	PM _{2.5} (satellite-based model, 2007-2015 mean)	Ghana
Ting	Statistical fragility and methodological constraints in metabolomic mediation of air pollution and cardiovascular risk	<i>JACC Adv</i>	Method review	Ambient PM (methodological commentary)	Not stated
Wittkopp et al.	Cleaner Air for Lower Cardiometabolic Risk: protocol for a double-blind, randomized, sham-controlled trial of HEPA filtration in adults with prediabetes	<i>Am Heart J</i>	Trial / intervention	PM _{2.5} (indoor, continuous)	USA
Zhang et al.	Acute impact of combined exposure to cold spells and heavy pollution on circulatory disease mortality in the Beijing-Tianjin-Hebei region	<i>Zhonghua Yu Fang Yi Xue Za Zhi</i>	Distributed-lag time-series	PM _{2.5} , heavy-pollution thresholds >75 / >115 / >150 µg/m ³	China
Reproductive & developmental					
Atalell et al.	Exposure to low-level fine particulate matter air pollution and developmental vulnerability at school entry: a population-based cohort study in Western Australia	<i>Environ Res</i>	Registry cohort + geospatial	PM _{2.5}	Australia
Chen et al.	Cord blood DNA methylation as a potential mediator between prenatal ambient air pollution and early infant allergic symptoms in a Shanghai birth cohort	<i>Pediatr Allergy Immunol</i>	Prospective birth cohort	PM _{2.5} , PM _{2.5} components (SO ₄ , NO ₃ , NH ₄), NO ₂	China
Hahad et al.	Air pollution and adverse birth outcomes: a narrative review of epidemiological and mechanistic findings	<i>J Environ Health Sci Eng</i>	Narrative review	PM _{2.5} , trimester-specific	Multi-region
Hung et al.	Late-pregnancy air pollution exposure and neonatal anthropometry with and without gestational hypertension	<i>iScience</i>	Prospective birth cohort	PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , CO, O ₃ (30 d pre-delivery, ambient monitors)	Taiwan
Wang et al.	Synergistic and antagonistic interactions of PM _{2.5} with high and low temperatures on sperm motility	<i>Environ Pollut</i>	Observational - cohort	PM _{2.5} with temperature, 0-90 / 0-9 / 10-14 d exposure windows	China
Respiratory & allergic					
Bogunia-Kubik et al.	Influence of environmental pollution on basophil reactivity to birch pollen allergens in atopic and non-atopic subjects	<i>Allergy</i>	Panel study	PM _{2.5} , PM ₁₀ (and O ₃)	Poland
Feng & Chao	Lower respiratory infection burden attributable to PM _{2.5} in China and the G20, 1990-2023	<i>Int J Biometeorol</i>	GBD secondary analysis	Ambient PM _{2.5} , PM _{2.5} household	China

First author	Focus	Journal	Design	Metric	Region
Govindaraju et al.	Protective role of dietary antioxidant intake on long-term effects of extreme PM _{2.5} exposure on respiratory health	<i>J Occup Environ Med</i>	Cross-sectional (cohort baseline)	Fire-sourced PM _{2.5}	Australia
Lancet Respir Med	The far-reaching effects of wildfire smoke	<i>Lancet Respir Med</i>	Metadata only (no abstract)	Wildfire PM _{2.5}	Global
Makrufardi et al.	Effect of ambient relative humidity and particulate matter on the levels of asthma therapy	<i>World Allergy Organ J</i>	Prospective cohort	PM _{2.5}	Taiwan
Ni et al.	Socioeconomic and regional influences on eosinophilic inflammation in chronic rhinosinusitis with nasal polyps	<i>World Allergy Organ J</i>	Retrospective clinical series	Annual mean PM _{2.5} (city-level)	China
Shi et al.	Air pollution and acute exacerbation of COPD in Hebei Province: a multi-city time-series study	<i>Medicine (Baltimore)</i>	Distributed-lag time-series	PM _{2.5} , PM ₁₀ , SO ₂ , NO ₂ , CO, 8-h O ₃	China
Wang et al.	COVID-19 social interventions were associated with amplified, not attenuated, indoor-source pollutant-respiratory associations in Xi'an	<i>Front Public Health</i>	Retrospective time series	PM _{2.5} , SO ₂ , NO ₂ , O ₃	China
Wu et al.	Air pollution and COPD-related health outcomes: bibliometric and visualisation analysis	<i>J Thorac Dis</i>	Mapping review	PM _{2.5} , PM ₁₀ , household PM	Global
Yang et al.	COPD and LRI burden attributable to household air pollution and second-hand smoke, 1990-2021	<i>J Thorac Dis</i>	GBD secondary analysis	Household PM from solid fuel	Global
Zhai & Meng	Impacts of apparent temperature and air pollutants on daily pneumonia hospitalizations and their interaction in rural areas of Western China	<i>Int J Environ Health Res</i>	Distributed-lag time-series	PM _{2.5} , PM ₁₀	China
Mechanistic toxicology					
Ayala et al.	Contrasting neurotoxic pathways triggered by PM ₁₀ in relation with organic molecular markers in suburban and rural sites in Catalonia	<i>Environ Pollut</i>	Exposure assessment + in vitro	PM ₁₀	Spain
Chen et al.	Gut microbiota signatures reflect particulate matter 2.5-exacerbated asthma in a murine model	<i>Allergy</i>	Animal model	PM _{2.5} (murine asthma model)	Taiwan
Cho et al.	Lactacaseibacillus paracasei ATG-E1 alleviates particulate matter-induced airway inflammation in association with altered inflammatory signalling, gut microbiota and faecal metabolites	<i>J Microbiol Biotechnol</i>	Animal model	PM ₁₀ D (PM ₁₀ + diesel exhaust particles)	East Asia
Choi et al.	Degradable PLGA/PCL membrane alveoli-on-a-chip for particulate-induced alveolar injury	<i>Biofabrication</i>	In vitro (cell lines)	Diesel particulate matter	In vitro
Gomes et al.	Senolytics reverse PM _{2.5} -induced cell and vascular dysfunction	<i>Cardiovasc Toxicol</i>	Animal + in vitro	Concentrated ambient PM _{2.5}	United States
Ichinose et al.	Time-dependent bronchus-associated lymphoid tissue formation and pulmonary inflammation induced by different particulate matter types	<i>J Hazard Mater</i>	Animal model	TiO ₂ nanoparticles, diesel exhaust particles, Asian sand dust	Japan
Mahmutovic Persson et al.	Chronic exposure to aerosolized PM _{2.5} in rats alters plasma protein levels and ex vivo coronary artery response to sphingosine-1-phosphate	<i>Environ Res</i>	Animal model	Re-aerosolized urban PM _{2.5} (acute high-dose vs chronic low-dose)	Europe
Seymore et al.	Gestational inhalation of nanoparticles alters placental glucose transporter expression and increases glucose flux in rats	<i>Toxicol Appl Pharmacol</i>	Animal model	TiO ₂ nanoparticles, 9.8 +/- 1.0 mg/m ³ whole-body inhalation	USA
Wei et al.	Seasonal source shifts drive oxidative potential of size-resolved particulate matter in coastal southeastern China	<i>J Hazard Mater</i>	Metadata only (no abstract)	Size-resolved PM oxidative potential	China
Yuan et al.	Hallmarks of lung cancer driven by inhalable particulate matter	<i>Innovation (Camb)</i>	Critical review	PM _{2.5} , PM ₁₀	Global
Zhang et al.	Impact of pleural anthracosis severity on structure and function of subcarinal lymph nodes	<i>J Thorac Dis</i>	Retrospective clinical series	Deposited carbonaceous particles	China
Occupational & indoor					
Alqahtani et al.	Biomarker-based assessment of tobacco smoke exposure in Saudi females via urinary cotinine	<i>PLoS One</i>	Repeated-measures biomarker	Tobacco smoke (urinary cotinine)	Saudi Arabia
Herkert et al.	Contribution of indoor air and dust to children's PFAS exposure in the Faroe Islands	<i>Environ Sci Technol</i>	Field measurement	Indoor air and settled dust, PFAS	Faroe Islands
Iavazzo et al.	Particle characterization in an urban synthetic turf: insights into turf and vehicular wear	<i>Environ Geochem Health</i>	Field measurement	Fine, submicron and nano-sized particles; Fe, Si, Al, Br, Ni, Ag, Rh	Italy

First author	Focus	Journal	Design	Metric	Region
O'Donnell et al.	Use of nicotine replacement therapy to reduce children's exposure to second-hand smoke in the home: a pilot randomised controlled trial in Scotland	<i>medRxiv (preprint)</i>	RCT process evaluation	PM _{2.5} (indoor)	Scotland
Ranjbar et al.	Evaluating diesel exhaust exposure in California homes using 1-nitropyrene in indoor air and dust	<i>Preprints.org (preprint)</i>	Measurement campaign	1-Nitropyrene, 2-nitropyrene, 2-nitrofluoranthene; continuous indoor black carbon	USA
Salcines et al.	Monitoring and risk assessment of airborne graphene oxide released during 3D printing with polylactic acid filaments	<i>J Environ Health Sci Eng</i>	Field measurement	Nanoparticle number conc. + respirable fraction	Spain
Shadmani et al.	A comprehensive health profile and exposure-risk analysis in an iron ore mining workforce	<i>J Environ Health Sci Eng</i>	Cross-sectional survey	Job-role dust exposure tertile (proxy)	Iran
Tan et al.	PM _{2.5} components, nutritional indices and physical activity vs lower eGFR in subway workers	<i>Nutrients</i>	Cross-sectional (cohort baseline)	PM _{2.5} component mixture	China
Zhao et al.	Occupational dust exposure and thyroid nodule screening: a predictive modelling approach for early detection in coal mining communities	<i>J Int Med Res</i>	Cross-sectional + prediction model	Occupational coal dust (unmeasured)	China
Sensing, forecasting & instrumentation					
Acton-Bond et al.	Rolling positive matrix factorisation for aerosol optical apportionment	<i>Atmos Environ</i>	Source-apportionment method	Absorption coefficient / optical apportionment	Europe
Adams et al.	Sampling sub-pixel scale variability in total column NO ₂ within a coastal urban environment	<i>Environ Sci Atmos</i>	Field campaign / remote sensing	Total column NO ₂	United States
Akaezuwu et al.	SFFO-RL intelligent environmental monitoring: smart sensing networks with artificial intelligence	<i>Environ Monit Assess</i>	Algorithm development	Unspecified pollutant concentration	Not stated
Albashir 2026	A physics-informed digital twin framework for health vigilance and indoor air quality decision support	<i>Res Sq (preprint)</i>	Simulation (system-level)	Indoor PM and IAQ composite	Not stated
Alexa et al.	Simultaneous determination of gaseous nitric acid and particulate nitrate in ambient air using continuous samplers and chemiluminescent online analysers	<i>Talanta</i>	Multi-instrument field evaluation	Gaseous HNO ₃ and particulate NO ₃ ⁻ , 1 s chemiluminescence	Czech Republic
Aparicio et al.	Sensor system design and deployment for environmental and visitor monitoring in cultural heritage sites	<i>Sensors (Basel)</i>	Network deployment + field evaluation	Low-cost pollution monitors + T/RH, VWC, wind; static and robot/drone-borne	Europe
Babu et al.	Capturing the invisible: hyperlocal variability of size-resolved particulate matter in a smart city	<i>Environ Monit Assess</i>	Mobile monitoring campaign	PM ₁ , PM _{2.5} , PM ₁₀ + size distribution	India
Bajaj & Settle	Rapid assessment of air pollution data quality using a composite index	<i>Environ Monit Assess</i>	Monitoring-record analysis	PM _{2.5}	USA
Balasubramanian et al.	Calibration model for PurpleAir-monitored PM _{2.5} across the United States	<i>Environ Pollut</i>	Calibration model comparison	PM ₁ , PM _{2.5} , PM ₁₀	United States
Benjamin et al.	Limitations of environmental extrapolation in low-cost carbon monoxide and PM _{2.5} sensors	<i>Sensors (Basel)</i>	Calibration model comparison	PM _{2.5} , CO	United States
Bobbia et al.	Multi-source dataset of NO ₂ and PM from mobile and fixed sensors in Rouen	<i>Data in Brief</i>	Network deployment / data descriptor	PM ₁₀ , PM _{2.5} , NO ₂ + RH/T/P	France
Cao et al.	On-road NO ₂ and particulate matter across urban road environments from city-scale mobile monitoring	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} , NO ₂	Not stated
Cho et al.	Greenspace, land use and PM _{2.5} across pandemic phases: a community scientist-led mobile sensing campaign	<i>Sci Total Environ</i>	Mobile monitoring campaign	PM _{2.5} (low-cost mobile sensors, street-level)	USA
Fawaz et al.	Surface cleaning of SAW-based microparticle sensors in a cascade impactor via SAW-induced droplet displacement	<i>Sensors (Basel)</i>	Instrument development	Size-segregated microparticles	France
Gao et al.	Feasibility of an air sensor array for real-time detection and characterisation of VOCs	<i>Atmos Meas Tech</i>	Instrument development	VOC sensor response (non-PM)	Laboratory
Gilardoni et al.	First continuous aerosol measurements at Testa Grigia at 3480 m a.s.l.: aerosol populations and dust transport dynamics in the southern European Alps	<i>Atmos Chem Phys</i>	Measurement campaign	Number size distribution + absorption coefficient, 1 h	Europe

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Habib et al.	Stable carbon isotopic composition of PM1 emissions from biomass and coal combustion under monitored combustion conditions	<i>Atmos Environ</i>	Metadata only (no abstract)	PM1, stable carbon isotopes	Not stated
Hsieh et al.	Identifying winter PM _{2.5} sources in an urban environment using receptor modeling and nonparametric trajectory analysis	<i>Aerosol Air Qual Res</i>	Metadata only (no abstract)	PM _{2.5} , source-resolved	Not stated
Kafi et al.	Evaluating the suitability of Sentinel-5P satellite data for air quality monitoring in Tehran	<i>Sci Total Environ</i>	Satellite retrieval evaluation	TROPOMI CO, NO ₂ , SO ₂ , O ₃ and UV Aerosol Index (no direct PM product)	Iran
Kirchhoff et al.	Evaluation of DMSO as working fluid in condensation particle counters	<i>Atmos Meas Tech</i>	Laboratory metrology	Ultrafine particle number	Laboratory
Laffolley et al.	A 3D-printed two-stage virtual impactor (uSPLIT) for radioactive aerosol size classification and direct analysis	<i>ACS Omega</i>	Instrument development	Aerodynamic diameter, 3 classes (>10, 1-10, <1 µm)	Japan
Li et al.	Spatiotemporally synchronized dual-wavelength photoacoustic-scattering measurement of aerosol optical properties	<i>Photoacoustics</i>	Laboratory metrology	Absorption & scattering coefficients, 450/532 nm	China
Liu et al.	Polarization calibration of spaceborne lidar using dense cirrus-scattered solar background	<i>Atmos Meas Tech</i>	Method / algorithm	Aerosol depolarization ratio	Global
Liu et al.	LRM-YOLO: lightweight YOLOv10n model for forest fire smoke detection in UAV images	<i>Sensors (Basel)</i>	Method / algorithm	Smoke plume (image proxy)	China
Perez-Pastor et al.	Robust quantification of anhydrosugars and reference material evaluation for biomass burning tracers	<i>Environ Sci Pollut Res Int</i>	Laboratory metrology	PM biomass burning tracers	Spain
Shirkhani et al.	Design and development of a high-throughput multipattern aerosol deposition device (MARTHA) for respiratory toxicity and health-risk assessment	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Aerosol deposition (unspecified)	USA
Wang et al.	Measuring heavy metals in atmospheric particulate matter: operational definitions, method comparability and environmental interpretation	<i>Atmos Environ</i>	Method review	Heavy-metal content of PM	Not stated
Wu & Qu	Interpretable short-term PM _{2.5} forecasting using meteorology-pollution coupling across Beijing stations	<i>Environ Monit Assess</i>	Method / algorithm	PM _{2.5}	Beijing
Yaqoob et al.	Semiparametric hybrid air quality forecasting using Prophet ensemble and feature selection	<i>Sci Rep</i>	Machine-learning model	PM _{2.5}	South Korea
Zhang et al.	High frequency of urban new particle formation on the Tibetan Plateau: quantifying formation rates, growth rates and CCN production	<i>Atmos Chem Phys</i>	Supersite observation	Particle number size distribution, CCN	China
Zhang et al.	Impact of aerosol-type assumption on Landsat 8 atmospheric correction over land	<i>Atmos Meas Tech</i>	Model intercomparison	Aerosol optical depth	Global
Zhang et al.	Air pollution transport from absorbing aerosol index across multiple payloads	<i>Atmos Environ</i>	Metadata only (no abstract)	Absorbing aerosol index	Global
Zhou et al.	Reliability-constrained identification of primary and elevated aerosol structures from a single tilted Mie lidar under dust-influenced conditions	<i>Atmos Environ</i>	Metadata only (no abstract)	Backscatter / aerosol layer structure	Not stated
d'Hondt et al.	Neural downscaling of air-quality simulations requires structural correction before spatial refinement	<i>Research Square (preprint)</i>	Deep-learning model	PM _{2.5} , PM ₁₀ , NO ₂ , O ₃	Europe
Exposure assessment & modelling					
Adeniji et al.	Long-term air quality and meteorological coupling dynamics in Abeokuta, Nigeria	<i>PLOS Glob Public Health</i>	Monitoring-record analysis	PM _{2.5} , PM ₁₀ + 4 gases, hourly	Nigeria
Chebaicheb et al.	A phenomenology of submicron organic aerosol components in urban France from long-term online observations and modelling	<i>Environ Int</i>	Source apportionment	PM1 organic aerosol, ACSM, rolling PMF (HOA/BBOA/COA/LO-OOA/MO-OOA)	France
Chen et al.	Inventory uncertainty and transboundary transport of biomass-burning black carbon in southern China from radiocarbon plus modelling	<i>Environ Sci Technol</i>	CTM simulation + source attribution	Black carbon, 14C source-resolved	China
Chillrud et al.	pcpr: principal component pursuit for exposure pattern recognition and outlier detection	<i>Environ Epidemiol</i>	Software / tool description	26 PM _{2.5} constituents	United States
Deng et al.	Impact of black carbon on daytime valley and slope winds in idealised WRF-Chem simulations	<i>Atmos Chem Phys</i>	Theory / simulation	Black carbon, radiative absorption	Idealised terrain

First author	Focus	Journal	Design	Metric	Region
Environ Sci Atmos	Multi-scale variability and lagged meteorological controls of PM _{2.5} in Delhi from wavelet coherence and causality analysis	Environ Sci Atmos	Monitoring-record analysis	PM _{2.5}	India
Fan et al.	Santa Ana winds control wildfire smoke transport and related health impacts in California during January 2025	Atmos Environ	Chemical transport modelling	Wildfire-sourced PM _{2.5}	North America
Florencio et al.	Comprehensive analysis of PM ₁₀ in the Metropolitan Area of Sao Paulo, Part I	Atmos Environ	Metadata only (no abstract)	PM ₁₀	Brazil
Gong et al.	Regional analysis of urban physical factors influencing air pollution: evidence from Seoul's living areas	Res Sq (preprint)	Spatial regression (GWR)	Composite AQ index (incl. PM _{2.5} , PM ₁₀)	East Asia
Gould & Jagarnath	Satellite monitoring for air quality and public health research in Africa: a scoping review	Research Square (preprint)	Scoping review	Satellite-derived PM _{2.5} and AOD	West Africa
Hao et al.	Source-specific black carbon responses to short-term interventions and cross-year emission changes in coastal Qingdao	Atmos Environ	Metadata only (no abstract)	Black carbon, source-resolved	China
Hernandez et al.	Urbanization and humidity impact atmospheric deposition of anthropogenic microparticles (including microplastics) in a humid subtropical city	Atmos Environ	Deposition sampling	Anthropogenic microparticle deposition flux	North America
Ibrahim	Air pollution status and temporal patterns in Bukit Rambai and Muar, 2013-2017	Research Square (preprint)	Ecological panel	PM ₁₀ , O ₃ , CO, SO ₂ , NO ₂	Malaysia
Li et al.	Upgrading diesel vehicle after-treatment technology significantly changed the composition of reactive nitrogen in exhaust	Environ Sci Technol	Vehicle emission experiment	Reactive nitrogen speciation (NO, NO ₂ , NH ₃ , N ₂ O)	China
Lin et al.	Elevated aerosol layers within the daytime mixed layer over ecologically sensitive areas of northwest China: diurnal variation, formation mechanisms and regional effects	Atmos Chem Phys	Measurement campaign	Aerosol vertical profile, 0.6-2 km layer	China
Lin et al.	Three-millennia alpine-lake records of divergent atmospheric Cd, Pb, Sb and Hg pollution in southwestern China	Environ Sci Technol	Environmental surveillance campaign	Atmospheric metal deposition: Cd, Pb, Sb, Hg	China
Makacha et al.	Unequal exposure: spatio-temporal and structural determinants of personal PM _{2.5} exposure in three sub-Saharan African geographies	Front Glob Womens Health	Observational exposure study	PM _{2.5}	The Gambia / Kenya / Mozambique
Park et al.	City-scale non-exhaust PM ₁₀ from emissions to exposure with a coupled chemistry-CFD framework	Environ Pollut	CTM simulation + source attribution	Non-exhaust PM ₁₀	Seoul
Ray et al.	Evaluating CCN activity through size-resolved hygroscopicity: synergistic observations and simulations in a tropical montane region	Atmos Environ	Metadata only (no abstract)	Size-resolved hygroscopicity (kappa), CCN	India
Stevens et al.	Biomass burning marker fingerprints for Australian hardwoods, softwoods and grasses	Atmos Environ	Metadata only (no abstract)	Biomass burning PM tracers	Australia
Sui et al.	Contrasting annual variation of organic and elemental carbon in the Pearl River Delta across the COVID-19 lockdown	J Environ Manage	Source apportionment	PM _{2.5} organic and elemental carbon, PMF	China
Yavuz et al.	High-resolution CFD modelling of traffic-related air pollutant dispersion in an urban area of Istanbul	Atmos Pollut Res	Metadata only (no abstract)	Traffic-related PM and NO _x	Turkiye
You et al.	Satellite-based estimation of ambient ammonia reveals meteorological influences and emission inventory uncertainties in South Korea	Environ Pollut	Satellite remote sensing	Ground-level NH ₃ (secondary PM _{2.5} precursor), CrIS, 15 km monthly	South Korea
Zhang et al.	National-scale heavy metal contamination in urban road dust across 82 Chinese cities	Environ Geochem Health	Review / synthesis	Road dust heavy metals	China
da Costa Cardoso et al.	Legacy Pb-Zn mine tailings as persistent sources of bioaccessible toxic elements in urban dust	Environ Monit Assess	Field measurement	Resuspended urban dust, Pb, Zn, Cd, Cu, Mn	Brazil
de la Torre et al.	A 15-year atmospheric record of intentionally produced persistent organic pollutants in Spain	J Hazard Mater	Environmental surveillance campaign	Semi-volatile POPs (passive PUF samplers)	Spain
Burden, policy & mitigation					
He et al.	Premature ischaemic stroke burden in adults aged 50-69 in China, India and the United States	Neuroepidemiology	GBD secondary analysis	Ambient PM _{2.5} (population attributable fraction)	Global

First author	Focus	Journal	Design	Metric	Region
Lee et al.	Toward a unified framework for climate and health research: gaps in exposure metrics and health outcomes	<i>J Korean Med Sci</i>	Mapping review	Climate exposure metrics incl. air pollution	Global
Li et al.	Drivers of air pollutant and greenhouse gas emissions since 2010 in the Greater Bay Area, China	<i>Atmos Environ</i>	Metadata only (no abstract)	PM and co-emitted pollutants, emissions	China
Liu D. et al.	COPD burden 1990-2050: joinpoint, age-period-cohort and Bayesian APC modelling of diverging risk-factor attribution	<i>Medicine (Baltimore)</i>	Burden estimation (GBD)	Ambient PM _{2.5} and household air pollution as GBD risk factors (PAF)	Global
Liu M. et al.	Divergent trends in near-surface ozone and nitrogen dioxide and ozone-attributable health burden in the Huaihe River Basin, 2013-2024	<i>Environ Monit Assess</i>	Satellite remote sensing	Satellite-derived surface MDA8 O ₃ and NO ₂ (no PM metric)	China
Liu et al.	Lung cancer incidence and mortality trends with age-period-cohort analysis, Zibo 2014-2024	<i>Front Oncol</i>	Ecological panel	Ambient air pollution (contextual)	China
Mou et al.	Impacts of ship emissions on regional air quality in the post-DECA era: the Yangtze River Basin	<i>Environ Pollut</i>	CTM simulation + source attribution	Ship-source PM _{2.5} , SO ₂ , NO _x , nitrate; WRF-Chem increments	China
Okeke et al.	Agricultural methane emissions and non-communicable disease mortality in West Africa, 2015-2020	<i>Environ Monit Assess</i>	Ecological panel, fixed effects	Agricultural CH ₄ (Mt CO ₂ e)	West Africa
Sathasivam & Veluchamy	Visual characteristics of PM from <i>Halymenia dilatata</i> microalgae biodiesel compared to low-sulfur diesel fuel	<i>Environ Sci Pollut Res</i>	Chamber emission characterisation	Gravimetric PM mass + SEM morphology	India
Suryanarayana Reddy et al.	Low-cost bio-based particle-capturing device for vehicular emissions: pilot study	<i>Environ Sci Pollut Res Int</i>	Bench / device experiment	PM _{2.5} , PM ₁₀ , CO, CO ₂	India
Tian et al.	Bioinspired honeycomb-structured nanofibrous membranes for high-efficiency PM capture	<i>Nanomaterials (Basel)</i>	Bench / device experiment	PM _{0.3} , PM _{2.5}	Laboratory
Urso et al.	Burden of dementia in Europe, 1990-2023: a systematic analysis from the Global Burden of Disease Study 2023	<i>Lancet Reg Health Eur</i>	Burden estimation (GBD)	Ambient PM (GBD risk-factor exposure)	EU-27
Vartanyan et al.	Structural inequities and chronic disease in Fresno County: a narrative review and multi-sectoral framework	<i>Cureus</i>	Narrative review	Air + pesticide pollution (qualitative)	North America
Wu & Cui	Impact of aircraft CO, NO ₂ and SO ₂ emissions on public health in Chinese airports	<i>J Glob Health</i>	Burden estimation (GBD)	CO, NO ₂ , SO ₂ (no PM metric)	China
Xie et al.	Thirty-year change and projection of Asian tracheal, bronchus and lung cancer burden	<i>J Thorac Dis</i>	GBD secondary analysis	Ambient particulate matter pollution	Asia
Other clinical endpoints					
Chin et al.	Impact of air quality and PM _{2.5} on hidradenitis suppurativa activity in the Sacramento region	<i>Res Sq (preprint)</i>	Retrospective time series	Days above 12 µg/m ³ PM _{2.5} (monthly count)	North America
Chow et al.	Blood metabolomic signatures linking air pollution to lung cancer	<i>Nat Commun</i>	Prospective cohort	PM _{2.5} , O ₃ , plus four co-pollutants	United States
Curr Hematol Malig Rep	Linking air pollution and hematologic malignancies: from epidemiology to molecular mechanisms	<i>Curr Hematol Malig Rep</i>	Mechanistic review	PM, benzene, NO ₂ , SO ₂ , arsenic, ethylene oxide	Global
Jain et al.	Ambient air pollution and glaucoma: systematic review and meta-analysis	<i>J Curr Glaucoma Pract</i>	Systematic review (PRISMA)	PM _{2.5} , PM ₁₀ , NO ₂ , NO _x , CO	Global
Li et al.	Global and national CKD burden attributable to fine particles: updated exposure-response meta-analysis	<i>Innovation (Camb)</i>	Umbrella review / meta-analysis	PM _{2.5}	Global
Liu et al.	Reply: statistical fragility and methodological constraints in metabolomic mediation	<i>JACC Adv</i>	Metadata only (no abstract)	PM _{2.5}	United States
Mandal et al.	Long-term exposure to ambient particulate matter and kidney function in urban Indian adults	<i>Kidney Int Rep</i>	Observational - cohort	Annual mean PM _{2.5} , 1 km x 1 km national spatiotemporal model at residential geocode	India
Negahi et al.	Climate-sensitive exposures and cancer outcomes: a systematic review and meta-analysis	<i>Int J Biometeorol</i>	Systematic review (PRISMA)	PM _{2.5} , wildfire smoke	Global
Schlaepfer et al.	Environmental hazards and risk of idiopathic urethral stricture	<i>Transl Androl Urol</i>	Retrospective cohort	Ambient PM, VOCs, metal oxides	Salt Lake City
Traisathit et al.	Air pollution and mortality in acute lymphoblastic leukaemia across age groups, northern Thailand	<i>PLoS One</i>	Retrospective cohort	PM _{2.5} , PM ₁₀ , O ₃ , CO	Thailand

First author	Focus	Journal	Design	Metric	Region
Wang et al.	Ambient PM _{2.5} and thyroid health: a cross-sectional study of 24,333 participants in Zhejiang, China	<i>J Hazard Mater</i>	Cross-sectional (multistage)	PM _{2.5}	China

Provenance and method

This rollup is an **aggregation only**. It pools `state/corpus/*.json`, `state/metrics.csv` and `state/trials.json` over 9 August 2026–15 August 2026 via `PMRW_START/PMRW_END` and **re-queries no literature API**, so every record was screened and deduplicated in the daily issue that first carried it. Per-record provenance, source-query windows, exclusions and known metadata defects are documented in those daily issues and are not restated here. **Daily issues covered: 9, 10, 11, 12, 13, 14 and 15 August 2026 — seven of seven, no missed runs and no gap in entry-date coverage. One exception to the no-re-query rule, made deliberately and recorded:** the 15 metadata-only DOIs were re-resolved against Crossref to test whether abstracts had since been deposited. **Zero of fifteen** had; six authorships were recovered and written back to the store (see “A label defect repaired by this rollup”). **Known defects carried into this rollup, stated rather than hidden:** (i) six endpoint labels meaning “no health endpoint” are unmapped in `ENDPOINT_CANON`, so `f2b` understates that bar by 30 records — the true figure is 65 of 134; (ii) journal names are free text and Research Square appears under two spellings, understating preprint share; (iii) 15 records carry no abstract from any checked source and are unassessed; (iv) the 11 August PDF as shipped shows journal names in place of six author names, corrected in the store but not in that PDF. Effect estimates are transcribed from abstracts and use heterogeneous exposure contrasts — see the forest-plot caption; **no meta-analytic summary is computed and none should be**. Trend and inflection statements are judgements over a seven-day window and are not statistical tests. Content derived from publisher metadata; abstracts remain the copyright of their respective publishers.